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PII: S2667-2952(26)00034-6
DOI: https://doi.org/10.1016/j.hcc.2026.100415
Reference: HCC 100415

To appear in: *High-Confidence Computing*

Received date : 18 March 2026
Revised date : 13 May 2026
Accepted date : 30 May 2026

Please cite this article as: R. Wang, J. Wang, Y. Huang et al., A survey on medical multimodal retrieval-augmented generation (RAG), *High-Confidence Computing* (2026), doi: https://doi.org/10.1016/j.hcc.2026.100415.

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A survey on medical multimodal retrieval-augmented generation (RAG)

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ARTICLE INFO

Keywords:
Large Language Models (LLMs)
Multimodal retrieval-augmented generation (RAG)
Medical AI
Clinical decision support

ABSTRACT

Large Language Models (LLMs) have demonstrated immense potential in the medical field, yet they face inherent limitations such as hallucination and knowledge obsolescence. Although retrieval-augmented generation (RAG) effectively mitigates these issues by incorporating external knowledge, traditional approaches are primarily limited to textual data, failing to leverage the rich multimodal nature of clinical information, which potentially compromises diagnostic accuracy. Multimodal RAG extends information processing capabilities to encompass diverse data types, thus overcoming the limitations of its text-only counterpart. This paper aims to provide a comprehensive survey of medical Multimodal RAG to facilitate a thorough understanding of its current landscape and future potential. Specifically, we introduce its related technical foundations, review the various types of multimodal data in the medical field, analyze its core challenges and corresponding solutions, explore the diverse applications and representative works of medical Multimodal RAG, and conclude by outlining evaluation methods and future research directions.

1. Introduction

In recent years, with the rapid advancement of AI and natural language processing (NLP) technologies, their applications in the medical domain have become increasingly diverse. The introduction of the Transformer architecture [1] and the subsequent emergence of Large Language Models (LLMs) have further accelerated the integration of AI into medical practice and research. With their robust capabilities in language understanding and generation, LLMs have demonstrated immense potential for application across various facets of the medical field. For instance, LLMs are capable of assisting in the identification of potential diseases by analyzing data such as patient symptoms and medical records and providing evidence-based recommendations for clinical decision-making through the analysis of extensive medical data [2]. LLMs can also automatically generate clinical notes and discharge summaries or extract key information from lengthy clinician-patient dialogues to produce medical summaries, thereby alleviating the documentation burden on clinicians [3]. Furthermore, LLMs can serve as powerful assistants in medical research by rapidly reviewing large volumes of literature, generating abstracts, and aiding in literature screening, thus significantly enhancing research efficiency [4].

The medical domain is fundamentally a safety-critical environment where computational systems must satisfy the stringent requirements of high-confidence computing. Despite the strong application value demonstrated by LLMs, their inherent limitations pose significant threats to clinical trust and patient safety. First, LLMs may produce hallucinations, generating information that appears plausible but is factually incorrect or fabricated [5]. In the medical domain, such hallucinations can lead to catastrophic consequences. Second, LLMs face the issue of knowledge obsolescence. Relying on static training data, the parametric knowledge of LLMs remains frozen once training is complete. As medical evidence evolves continuously, some of the knowledge memorized by models often becomes outdated, which may result in the provision of harmful medical advice or failures in clinical reasoning [6]. Furthermore, functioning as black-box models, LLMs often lack transparency and interpretability in their reasoning processes [7]. This consequently hinders users from effectively assessing the reliability of generated content, thereby impeding the models' deeper integration into clinical decision-making. To address the above issues, retrieval-augmented generation (RAG) has been introduced as an effective strategy. RAG is a paradigm that integrates generative models with external databases. Before generating a response, a retriever first identifies relevant document fragments from external knowledge sources based on the input; subsequently, a generator concatenates this retrieved information with the original input, utilizing it as enriched context to guide the generation process [8]. Its primary advantage lies in the ability to synergistically combine the intrinsic knowledge of LLMs with vast, dynamic external knowledge bases. By outputting content based on retrieved external knowledge and providing traceable references, RAG reduces factual inaccuracies and enhances the verifiability and overall trustworthiness of the responses. In medical applications, by integrating clinical guidelines and literature resources into the response generation process, RAG can significantly improve the accuracy of diagnostic suggestions and preoperative assessments, while reducing the output of false information [9, 10].

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However, traditional RAG techniques also possess limitations, particularly when addressing the challenges of multimodal data in the medical domain. Clinical medical data is inherently multimodal, and modern medical decision-making often necessitates the comprehensive analysis of data in various forms, including text (e.g., clinical notes, radiology reports), images (e.g., X-rays, CT scans), and structured data (e.g., laboratory results, vital signs) [11]. Traditional RAG systems are designed primarily for pure text. In medical scenarios where high-confidence decision-making relies on the corroboration of multi-source evidence, discarding non-textual modalities introduces significant uncertainty and compromises the safety and reliability of the generated content [12]. Multimodal RAG emerges as a critical paradigm to overcome this fundamental limitation, extending the traditional RAG paradigm from retrieving only textual knowledge to a new model that can retrieve information from diverse sources, such as medical images and physiological signals. The core significance of this approach lies in providing a richer and more comprehensive context for the generation process by introducing and fusing external knowledge from multiple modalities and improving the quality, accuracy, and trustworthiness of the final output by leveraging complementary cross-modal cues [13]. From a technical perspective, realizing this potential necessitates addressing challenges across key dimensions, such as aligning multimodal representations within a unified space, effectively fusing heterogeneous knowledge sources, and controllably integrating retrieved evidence into clinical reasoning. These dimensions jointly define the methodological landscape of medical Multimodal RAG and provide a systematic lens for understanding recent advances. Consequently, developing Multimodal RAG techniques tailored to the medical field in order to bridge multimodal medical data and enhance the reliability of LLMs in medical scenarios represents a research issue of significant value.

Driven by the imperative of high-confidence computing in healthcare, this paper aims to provide a comprehensive survey of the emerging field of medical Multimodal RAG. It systematically examines the paradigm's technical foundations, multimodal data types, key challenges, applications, representative frameworks, performance evaluations, and future directions, with the goal of offering a clear reference framework for researchers and practitioners in the field. The organizational structure of this survey is illustrated in Figure 1. In the rest of this paper, Section 2 introduces the technical foundations of Multimodal RAG. Section 3 reviews the various types of multimodal data used in medicine and discusses their inherent challenges. Section 4 provides a detailed discussion of the key challenges of Multimodal RAG and their potential solutions, addressing aspects such as semantics, data, trustworthiness, domain adaptation, complex reasoning, and deployment. Section 5 explores the diverse applications of Multimodal RAG in the medical domain, including clinical diagnostic assistance, medical question answering, automated report generation, education, and personalized treatment planning. Section 6 surveys representative research works on medical Multimodal RAG, categorizing them into three dimensions: retrieval control, integration of structured and factual knowledge, and innovation in paradigms. Section 7 provides an overview of commonly used evaluation datasets and metrics. Section 8 evaluates and compares the performance of representative frameworks across different clinical tasks. Section 9 discusses the future research directions for medical Multimodal RAG. Finally, Section 10 concludes the paper by summarizing the current state and future prospects of Multimodal RAG in the medical domain.

2. Related technical foundations

To facilitate a thorough understanding of medical Multimodal RAG, this section introduces its technical pillars. We begin by outlining the fundamental principles and core modules of RAG systems. Subsequently, we explore essential techniques for multimodal representation learning, focusing on data fusion and alignment. Finally, we review the evolution from unimodal to multimodal paradigms and introduce representative medical foundation models.

2.1. Fundamental principles of RAG

The foundational paradigm of RAG, as proposed by Lewis et al. [8], combines parametric and non-parametric memory. Parametric memory is embodied by a pre-trained LLM, where all knowledge learned during training—including linguistic patterns, factual information, and reasoning abilities—is stored within its neural network weights. This knowledge is inherent and static, and updating it requires a costly retraining process. Non-parametric memory, on the other hand, consists of an external, easily accessible knowledge base that is simple to update. This memory is accessed through a dedicated retriever. The RAG workflow typically follows a "retrieve-then-read" pattern. When a user poses a query, the retriever first identifies the most relevant information snippets from the non-parametric knowledge base. This retrieved information, along with the original query, is then supplied as an augmented prompt to the parametric generator. Subsequently, the generator synthesizes the final answer based on this real-time, verifiable information.

2.1.1. The retrieval module

The retrieval module is the information acquisition engine of a RAG system. Its core task is to efficiently and accurately locate the most relevant information snippets from a vast knowledge base based on the user's query. The quality of retrieval directly determines the quality of the context available to the generator, thereby influencing the accuracy of the final answer.

Fundamental retrieval methods include sparse retrieval, dense retrieval, and hybrid search. Sparse retrieval mainly includes algorithms like TF-IDF [14] and BM25 [15]. They operate on the principle of keyword matching, assigning higher weights to query terms that appear frequently in a document but are relatively rare in the overall corpus.

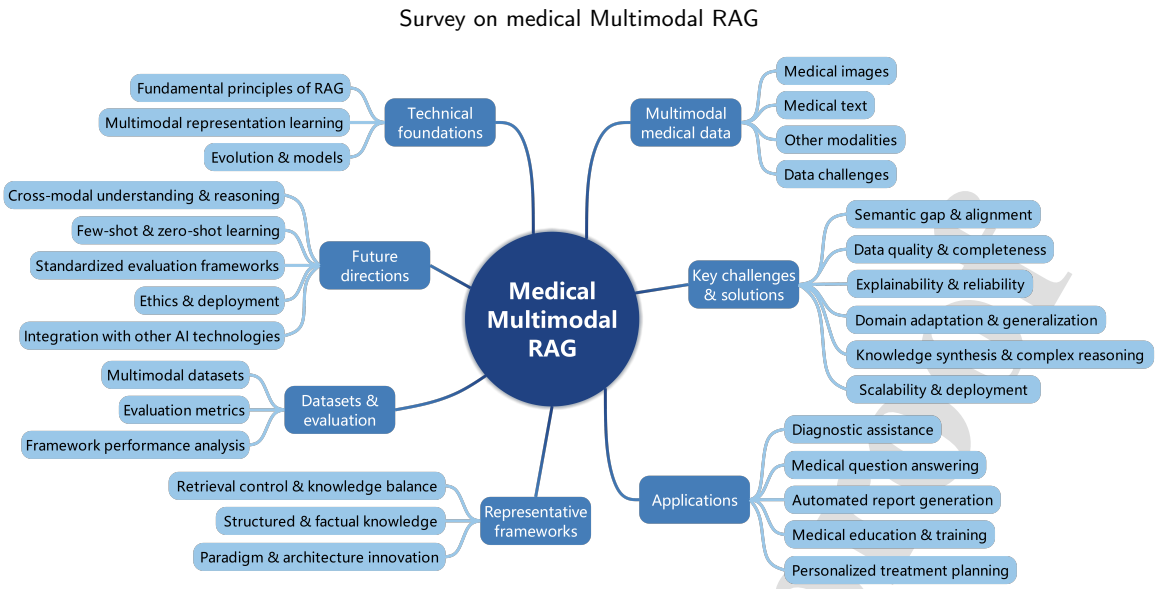


Figure 1: Organization of this survey. The mind map outlines the key topics and logical flow presented in this paper regarding medical Multimodal RAG.

The BM25 algorithm also takes document length into account. Sparse retrieval excels at handling queries containing specific technical terms, abbreviations, or entity names but has limited capabilities in understanding conceptual or synonymous queries. Dense retrieval typically utilizes encoder models (e.g., BERT [16]) to encode queries and document chunks into embeddings within the same vector space. The retrieval process is then completed by calculating the similarity between the query and document vectors. The core advantage of dense retrieval lies in its ability to capture semantic meaning, allowing for successful matches even when the query and document do not share keywords, as long as their meanings are similar [17]. However, the upper bound of its effectiveness is limited by the knowledge contained within the pre-trained retriever model itself. To combine the semantic understanding of dense retrieval with the lexical precision of sparse retrieval, hybrid search emerged. It executes both dense and sparse retrieval simultaneously, then merges and re-ranks the results using a weighted fusion strategy. This method effectively balances the needs for semantic understanding and precise keyword matching, making it suitable for scenarios that require both interpreting user intent and accurately retrieving proper nouns [18].

As RAG applications have deepened, researchers have found that a simple, single-pass retrieval is often insufficient for complex scenarios. This limitation is particularly pronounced in the medical domain, where answering clinical queries frequently requires synthesizing information across lengthy patient records, navigating complex medical guidelines, or constructing multi-hop evidence chains. Consequently, a series of advanced retrieval techniques have been developed, such as the FLARE and RAPTOR. Traditional RAG performs retrieval only once before generation begins, which is inadequate for long-form generation tasks that require continuous information. The FLARE framework, proposed by Jiang et al. [19], transforms retrieval into an active, iterative process. It first generates a temporary future sentence and then checks for the presence of low-confidence tokens. If such tokens exist, FLARE uses this predicted future content as a query to retrieve additional relevant information and then regenerates the sentence with this new information. This on-demand retrieval strategy avoids unnecessary information acquisition, making the generation process more dynamic and factually reliable. Traditional RAG typically retrieves isolated text chunks, which limits a holistic understanding of the overall context of long documents. The RAPTOR model, proposed by Sarthi et al. [20], addresses this by recursively embedding, clustering, and summarizing text chunks to build a hierarchical tree structure from the bottom up. At inference time, the model can retrieve from different levels of this tree—from leaf nodes containing specific details to higher-level nodes containing generalized summaries. This enables the model to integrate and understand information at different levels of abstraction across an entire long document, achieving significant performance improvements on tasks that require complex, multi-step reasoning.

2.1.2. The generation module

The generator, which is the LLM (or MLLM) itself, serves as the core reasoning component of the RAG framework, responsible for producing coherent and accurate responses. In practice, the retrieved document chunks are concatenated with the original query to form an augmented prompt, which guides the model's generation process.

The performance of the generator is highly dependent on the quantity and quality of the retrieved contexts. Research has found that as the number of retrieved snippets increases, system performance steadily improves, but this trend often stagnates or even declines after reaching a certain number of snippets. This suggests that the model reaches a state of saturation due to an overabundance of information

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[21]. Furthermore, an excessive amount of retrieved contexts may contain information that is irrelevant or noisy, which adversely impacts the generator's ability to produce a response. To address the issue of irrelevant retrieved information potentially degrading output quality, the MMed-RAG framework proposed by Xia et al. [22] introduces an adaptive context selection strategy, which dynamically controls the amount of retrieved context by evaluating similarity scores. The CAG architecture, proposed by Heydari et al. [23], uses a lightweight mechanism to dynamically determine whether a query truly needs external context retrieval before invoking the full RAG pipeline, thereby avoiding unnecessary retrieval, improving efficiency, and preventing noise from entering the system at the source.

Furthermore, simply concatenating a retriever and a generator often fails to achieve optimal performance. More advanced methods focus on fine-tuning these two components so they can work more synergistically. The RA-DIT framework, proposed by Lin et al. [24], is a lightweight fine-tuning method that first applies instruction tuning to the generator, training it on tasks that require utilizing a given document to answer, thereby teaching it how to better integrate and utilize retrieved information; subsequently, the retriever is fine-tuned to return results that are more aligned with the language model's preferences. This step aligns the retriever's ranking criteria with the generator's actual needs, creating a more coordinated and consistent system.

2.2. Multimodal representation learning

2.2.1. Methods for multimodal data fusion

In modern clinical practice, the decision-making process is inherently multimodal, potentially integrating information from various sources such as radiological images, EHRs, pathology reports, physiological signals, and genomic sequences. Developing AI systems capable of integrating and understanding these heterogeneous data streams is crucial, and multimodal representation learning forms the technical cornerstone of such systems. To navigate these techniques effectively, it is essential to explicitly distinguish between two foundational concepts: *fusion* and *alignment*. Specifically, fusion describes *when* information from different modalities is combined within a model's architecture (e.g., at early, intermediate, or late stages). In contrast, alignment (discussed in Section 2.2.2) refers to *how* different modalities are mapped into a comparable semantic space to enable cross-modal understanding. Focusing first on fusion, methods are generally categorized based on the architectural stage of integration.

Early fusion, also known as feature-level fusion, is centered on the idea of merging raw data or shallow features from different modalities at the initial stages of model training to form a unified feature vector. This vector is then fed into a single downstream model for processing. Common techniques for achieving this fusion include simple feature concatenation, pooling, or the use of gated units [25]. The primary advantage of early fusion is its ability to capture correlations and interactions between low-level features of different modalities early in the network. However, it can suffer from the modality gap and is prone to creating extremely high-dimensional features after concatenation, leading to the curse of dimensionality.

Late fusion, or decision-level fusion, first trains a dedicated model for each modality independently. After each model produces its own predictions or high-level representations, these outputs are integrated at the decision level to form a final conclusion. Common techniques for integrating these independent predictions include averaging or training a meta-learner to weight the predictions from each modality [25]. Late fusion offers greater architectural flexibility by allowing for the design of specialized, optimized processing methods for each data type. The main drawback of this approach is that because fusion occurs after feature extraction for each modality is complete, it cannot leverage potential correlations between low-level features across modalities, which may result in the loss of critical cross-modal information.

Intermediate fusion represents a compromise between early and late fusion. It initially processes data from each modality through separate models to extract features. These features are then merged and input into a unified downstream model for the final prediction. The main advantage of intermediate fusion is that it integrates feature representations from different modalities at the middle layers of the neural network, thereby enabling the capture and integration of complex cross-modal interactions at various levels of abstraction. However, its model architecture can become very complex and typically requires a large amount of data for effective training. Consequently, this method excels in applications that provide rich data and require deep feature interaction, such as combining pathological images with genomic data for cancer prediction [26].

2.2.2. Multimodal embedding and alignment techniques

The goal of multimodal embedding and alignment techniques extends beyond simple data stream merging, aiming to create a unified and semantically coherent shared embedding space where data from different modalities can be effectively compared, retrieved, and interacted with. The core of this approach lies in establishing semantic relationships across modalities, ensuring that representations from different modalities are aligned within the same common space. This is a technical prerequisite for achieving advanced tasks such as cross-modal retrieval and visual question answering [27].

Inspired by the tremendous success of the CLIP [28] model in the general domain, medical Vision-Language Models (VLMs) widely adopt contrastive learning to align the representations of paired images and text reports. Contrastive learning is a self-supervised learning method whose core idea is to pull semantically similar samples (positive pairs) closer in the embedding space while pushing dissimilar samples (negative pairs) apart [29]. In multimodal scenarios, a typical positive pair consists of an image and its

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corresponding textual description. This contrastive learning objective is typically achieved using a dual-encoder architecture. An image encoder (such as ResNet or a Vision Transformer) is responsible for converting input images into visual embeddings, while a text encoder (such as BERT) converts text descriptions into text embeddings. These two encoders process their respective modalities separately and project their outputs into a shared representation space of the same dimension. The model is optimized via a contrastive loss function, which aims to maximize the similarity between the embedding vectors of positive pairs while minimizing their similarity with all other negative pairs within a batch.

Alignment techniques in the medical field have evolved from simple global alignment to more complex, fine-grained, and multi-level alignment to capture the subtle features in clinical data. Global instance-level alignment is the most fundamental alignment method. For example, the ConVIRT model treats each image-report pair as a single instance, learning representations using a ResNet-based image encoder and a BERT-based text encoder. Through a bidirectional contrastive objective, the model maximizes the similarity between paired image-text representations, thereby learning high-quality visual and text representations [30]. However, this method may overlook the precise correspondence between local lesions in an image and specific descriptions in the report. To address the limitations of global alignment, researchers have proposed methods for alignment at a finer granularity. For instance, the LoVT model proposed by Müller et al. [31] innovatively segments the image into multiple patches and the report into multiple sentences, then uses an attention mechanism to explicitly align the representations of corresponding regions and sentences. This local contrastive loss encourages the model to learn representations that are more sensitive to tasks such as lesion localization in specific image regions. The PRIOR framework proposed by Cheng et al. [32] goes a step further by not only performing global and local alignment but also constructing a prototype memory bank for sentence embeddings from the reports. This guides the language features to focus on high-level, discrete concepts closely related to medical images, thereby learning more representative features. Furthermore, the MACL framework proposed by Zeng et al. [33] introduces a multi-level asymmetric contrastive learning strategy that performs contrastive learning at the feature, image, and pixel levels simultaneously, ensuring that both the encoder and decoder can capture a wide range of information from coarse to fine during the pre-training phase.

2.3. Evolution of the RAG framework

2.3.1. From Unimodal RAG to Multimodal RAG

Unimodal RAG, or traditional RAG, is a framework designed to enhance the performance of LLMs by integrating external, text-only knowledge [8]. Its core principle involves utilizing a retriever to find relevant information snippets from a large-scale text corpus, such as medical literature or clinical guidelines, before generating an answer. Subsequently, these retrieved text segments, along with the original query, are fed into a generator (an LLM), which then produces the final, factually grounded response.

Although Unimodal RAG is effective in integrating textual knowledge, the inherent limitations of its architecture are particularly pronounced in the highly complex domain of medicine. Clinical practice is a typical multimodal information processing procedure, where physician diagnosis and decision-making heavily rely on the comprehensive interpretation of various types of data, such as radiological images, electrocardiogram signals, and structured laboratory results. A text-only RAG system cannot directly utilize this non-textual information, a limitation that leads to incomplete contextual understanding and poses a potential risk to patient safety [34]. For instance, in diagnostics, a radiologist's textual report is an interpretation and summary of a medical image, but the image itself constitutes the primary and most critical diagnostic evidence. A traditional RAG system can only process this text report and is unable to directly analyze the raw, richer information contained within the image. This reliance on second-hand information not only results in the loss of critical visual details but may also be misleading due to the subjectivity or imprecision of the textual description, thereby compromising diagnostic accuracy.

Multimodal RAG is a framework extended from the traditional RAG paradigm. Its core mechanism lies in the fact that the system is no longer confined to a single data modality; instead, it utilizes a retriever to identify and acquire knowledge fragments relevant to the user query from a multimodal knowledge base. Subsequently, this retrieved multimodal contextual information is integrated and fed into a generative model, thereby producing a comprehensive response grounded in both the original query and the external multimodal evidence [13]. Multimodal RAG represents a significant evolution of the RAG framework, primarily driven by the need to overcome the limitations of text-only RAG. By integrating and leveraging multi-source, multimodal information, it equips the generative model with a more comprehensive and contextually rich knowledge background, thereby enhancing the quality and factual accuracy of the generated outputs.

The transition from Unimodal RAG to Multimodal RAG necessitates fundamental upgrades and extensions to the system architecture, which typically include: (1) replacing the original text-only encoder with a powerful multimodal encoder capable of mapping data from different modalities into a unified, shared vector representation space; (2) substituting the original text-only knowledge base with a composite one that stores multimodal data, the construction and indexing of which is foundational to the Multimodal RAG system; and (3) optionally upgrading generators from standard LLMs to Multimodal Large Language Models (MLLMs) that can directly process multimodal inputs. This facilitates comprehensive cross-modal reasoning to ultimately generate an

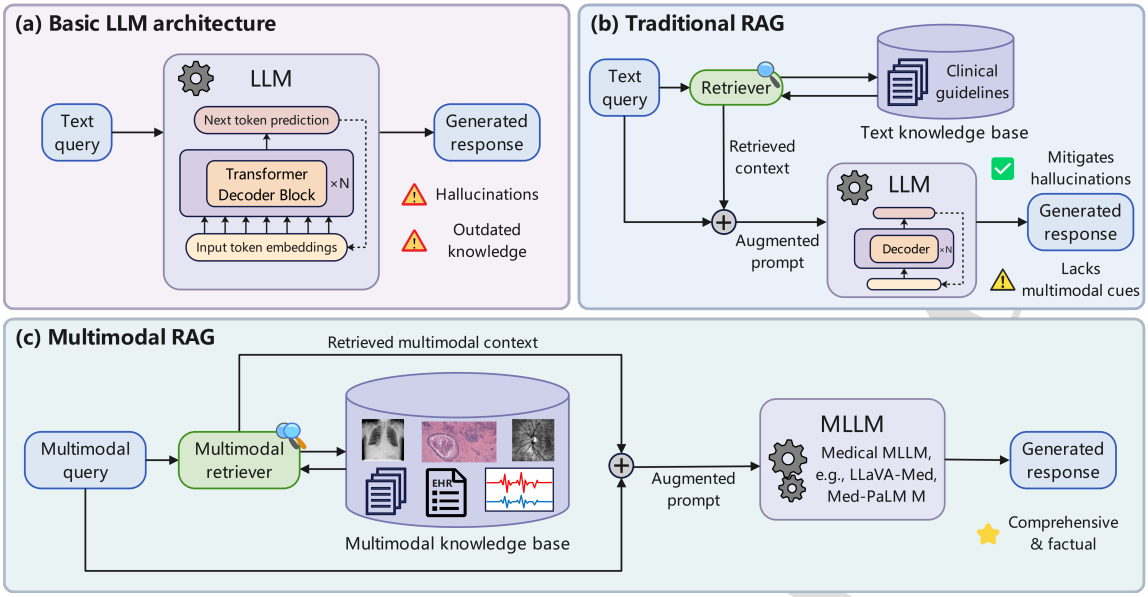


Figure 2: Architectural evolution from standard LLMs and traditional RAG to Multimodal RAG.

answer that integrates all information [35]. To intuitively illustrate this transition, Figure 2 compares the basic architectures of standard LLMs, traditional RAG, and Multimodal RAG.

2.3.2. Representative medical models

Medical LLMs: Medical LLMs are developed by incorporating medical domain knowledge into general-purpose LLMs through specialized fine-tuning and can serve as the core reasoning component in medical RAG systems [36]. Ideally, these models possess a profound understanding of concepts such as clinical terminology, symptoms, and diseases. Compared to non-fine-tuned general-purpose models, they are capable of delivering more accurate and reliable responses within medical contexts. Medical LLMs have been widely applied in assisting medical diagnosis and providing treatment recommendations, while playing a pivotal role in areas such as automated clinical documentation generation (e.g., clinical notes and discharge summaries), medical education, and mental health services [37, 38].

For instance, Med-PaLM 2 [39] is an LLM designed to provide high-quality medical question answering. The model, based on PaLM 2, is subsequently instruction fine-tuned on a mixed training set of multiple medical question-answering datasets from the MultiMedQA benchmark. It also introduces an innovative prompting strategy, ensemble refinement, during its training and inference processes. Med-PaLM 2 achieves an accuracy of up to 86.5% on the challenging MedQA (USMLE) dataset, reaching an expert level for the first time. For the task of generating long-form answers to consumer medical questions, the model demonstrates the potential to surpass physician-written answers in human expert evaluations. Me-LLaMA [40] is an open-source medical LLM family based on Llama 2. The model is first continuously pre-trained on a meticulously designed, large-scale mixed dataset and then fine-tuned on a high-quality, diverse medical instruction dataset. On a novel and comprehensive medical evaluation benchmark named MIBE, Me-LLaMA achieves better overall performance than all then-existing open-source medical LLMs across various scenarios, including zero-shot, few-shot, and supervised learning. It also challenges the standing of top-tier closed-source models on multiple tasks. Simultaneously, its unique continuous pre-training strategy successfully mitigates the catastrophic forgetting problem.

Medical MLLMs: Medical MLLMs are designed to simultaneously process multiple forms of data, such as medical text and visual information, to accommodate the inherent multimodal characteristics of the medical field. By integrating language models with visual encoders, medical MLLMs are capable of jointly analyzing diverse input sources, such as electronic health records and radiological images. Compared to text-only models, medical MLLMs can provide more comprehensive analyses, thereby facilitating more accurate diagnoses and treatment decisions [41]. Their specific application scenarios cover multiple aspects, including multimodal medical diagnosis, radiology report generation, and surgical assistance [37, 42].

In the development of medical MLLMs, LLaVA-Med and Med-PaLM M stand out as two representative works. LLaVA-Med [43] is an open-source biomedical visual-language conversational model whose architecture follows the LLaVA framework. Its main components include a pre-trained visual encoder, a large language model, and a trainable linear projection layer. Researchers utilized an innovative self-instruct method with GPT-4 to transform simple image-text data into high-quality conversational training

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data, achieving fast and efficient training through a two-stage curriculum learning approach. The model is capable of answering open-ended questions about image details and providing relevant explanations, demonstrating strong VQA performance. Med-PaLM M [44] represents a groundbreaking step toward generalist biomedical AI. Through multi-task fine-tuning on the PaLM-E multimodal architecture using the specially constructed and comprehensive MultiMedBench benchmark, it successfully creates a generalist model capable of processing diverse data types such as text, imaging, and genomics with a single set of weights. On all 14 tasks in MultiMedBench, Med-PaLM M achieved state-of-the-art (SOTA) or competitive performance at the time, outperforming specialized models on many of the tasks.

3. Multimodal data in medicine

Medical decision-making relies on diverse information sources. This section categorizes the primary types of multimodal data utilized in medical RAG, ranging from radiological images and clinical texts to genomics and time-series signals. Furthermore, we analyze the inherent characteristics of these data types and discuss the fundamental challenges they present, including heterogeneity, privacy concerns, and data scarcity, which significantly influence system design.

3.1. Medical image data

X-ray radiography is a technique that uses high-energy electromagnetic waves to penetrate the human body, generating an image based on the differential absorption of the rays by various tissues, such as bone and soft tissue. This technology is primarily used to diagnose fractures, detect lung pathologies, visualize the structure of the digestive tract, and guide medical procedures like catheter insertion [45]. LLMs can serve as clinical tools by generating corresponding reports based on X-ray images, potentially reducing diagnostic workloads and improving radiologist efficiency [46, 47, 48].

CT is an advanced X-ray imaging technology that acquires cross-sectional data of the body using a rotating X-ray emitter and detector, which a computer then reconstructs into detailed tomographic images. It serves to diagnose diseases and trauma in numerous body parts and is crucial for planning treatments, guiding interventional procedures, and monitoring the effectiveness of therapies, such as for cancer [45]. MLLMs have reported their primary success in single-slice CT report generation [43, 44]. Moreover, with appropriate adaptation and development, they can also interpret 3D CT scans to automatically generate comprehensive radiology reports and answer questions about specific anatomical regions, demonstrating their potential to function as a collaborative assistant in clinical diagnosis [49, 50].

Magnetic resonance imaging (MRI) is a technology that utilizes strong magnetic fields and radiofrequency waves to generate signals from protons within the body, which are then received by a computer to construct detailed images. This technique is widely applied to examine abnormalities of the brain, spinal cord, joints, and abdominal organs, while also providing support for surgical planning [45]. A fine-tuned LLM can serve as a diagnostic aid, assisting radiologists with brain MRI differential diagnosis by processing their text descriptions of case information and imaging findings [51]. It automatically classifies written brain MRI reports, identifies potential abnormalities, and efficiently selects cases for research purposes [52, 53].

Ultrasonography generates real-time, dynamic images by transmitting high-frequency sound waves into the body via a transducer and capturing the echoes that reflect back from internal organs. This technique is commonly used to monitor fetal development, visualize organs in the abdomen and neck, guide needle injections, and assess cardiac function [45]. While LLMs can be prompted to directly analyze and classify ultrasound images, their current performance is limited and falls short of that of human experts [54]. However, when textual descriptions of ultrasound images provided by radiologists are available, their diagnostic capabilities are significantly enhanced, indicating their potential value as an auxiliary diagnostic tool [55, 56].

Anterior segment optical coherence tomography (AS-OCT) is a non-contact, high-resolution imaging modality widely utilized in ophthalmology for visualizing the anterior structures of the eye. It plays an important role in the clinical diagnosis and severity grading of ocular diseases, particularly for tasks such as nuclear cataract classification [57, 58]. In recent years, advanced machine learning techniques and recalibration networks have been developed to address specific challenges in ophthalmic image analysis, including imbalanced clinical data distributions and region-specific feature extraction [59, 60]. Within the context of Multimodal RAG systems, integrating these highly specialized ophthalmic images with relevant clinical textual records represents a promising direction. MLLMs augmented with specialized ocular imaging knowledge have the potential to provide more reliable ophthalmic diagnostic assistance.

3.2. Medical text data

Electronic Health Records (EHRs), also known as Electronic Medical Records (EMRs), are comprehensive digital files of a patient's personal medical history and health information. These records contain not only structured data such as demographics, lab results, and vital signs, but also encompass a large volume of unstructured text like clinical notes, discharge summaries, and radiology reports. The primary use of EHRs is to enhance healthcare services by facilitating clinical decision-making and quality management, as well as to provide data support for scientific research in the biomedical field [61]. LLMs can analyze EHRs to perform tasks such as clinical data extraction [62, 63], text summarization [64], information classification [65, 66], patient-provider dialogues [67], and disease risk prediction [68, 69], ultimately aiming to enhance diagnostic efficiency and treatment outcomes by assisting in clinical decision-making, reducing documentation workload, and improving patient-provider communication.

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Clinical notes are detailed texts written by healthcare professionals to document a patient's condition, diagnosis, and treatment process, serving as a vital component for treatment planning and medical research. Preprocessing physician notes using LLMs can significantly improve the performance of downstream information extraction systems by enhancing the quality of the note's text [70]. LLMs can efficiently automate the extraction of key structured information from clinical notes, such as identifying specific pathological classifications from pathology reports [71]. In addition, leveraging LLMs to automatically augment clinical notes—by simplifying the text, defining terms, extracting key information, and generating frequently asked questions (FAQs)—has shown a significant positive impact on patients' reading and comprehension [72]. These capabilities can significantly reduce the documentation burden on physicians, allowing them to dedicate more time to direct patient care, thereby enhancing clinical efficiency and the quality of healthcare services. A discharge summary is a document provided to patients upon leaving the hospital. It explains their treatment, provides instructions for follow-up care, and acts as a key link in ensuring continuity of care and enhancing patient self-management. LLMs can transform original discharge summaries, which are often filled with technical jargon, into patient-friendly language and formats. Summaries transformed by the model show significantly improved readability and understandability [73], which helps patients better comprehend their health status and medical advice, potentially leading to better treatment adherence. Other unstructured medical records, such as patients' free-text symptom descriptions or medical dialogue transcripts, also contain rich clinical information. Studies have found that LLMs can accurately extract structured symptom labels from patients' free-text narratives with high specificity and sensitivity [74]. Summaries of medical dialogues generated by LLMs have been favored by human experts for their readability and overall quality [75].

Medical literature comprises publications that document academic research and clinical findings, primarily including scholarly articles from databases such as PubMed, Web of Science, and medRxiv. These publications serve as the cornerstone of evidence-based medicine, typically featuring rigorous study designs, data, and conclusions. They represent a core informational resource for advancing biomedical knowledge and informing clinical practice. LLMs build their foundational expertise in the medical domain and enhance their ability to accurately process and respond to medical information through continual pre-training on hundreds of billions of tokens from medical literature, as well as instruction tuning using literature-based tasks [40]. These models can efficiently process vast amounts of medical literature, assisting researchers with literature retrieval, abstract writing, and outline preparation tasks [76]. This can significantly improve the efficiency with which researchers screen and evaluate massive volumes of literature, helping medical experts and scientists to more rapidly comprehend and apply the latest biomedical information and accelerating knowledge-intensive tasks like systematic reviews [77].

3.3. Medical data from other modalities

Genomic data constitute a multi-layered information system that refers to the DNA sequence information within an individual's cells, ranging from single genes to the complete genome obtained through whole genome sequencing or whole exome sequencing. This concept is often extended to encompass downstream molecular derivatives such as RNA (transcriptomics), proteins (proteomics), and metabolites (metabolomics) [78]. In essence, genomic data represent the complete set of instructions necessary for the construction and maintenance of a living organism. The analysis of this data is essential for understanding the genetic determinants of health, identifying disease risks, and developing personalized therapies. LLMs are increasingly being adapted to interpret the complex, language-like structures within genomic data, from raw DNA sequences to gene sets derived from omics experiments [79, 80]. By analyzing this data, models can identify functional genetic variants, infer gene-regulatory networks, and discover the common biological functions of gene sets [81, 80]. LLMs are also capable of interpreting genetic variant annotations and clinical text to aid in variant classification and to prioritize Variants of Uncertain Significance (VUS) for expert review, thereby enabling more patients to benefit from personalized genomic medicine [82, 83].

Medical time series data are sequential observations collected over time that measure health-related physiological signals and vital signs, such as electroencephalography (EEG), electrocardiography (ECG), and intensive care unit (ICU) readings [84]. These data are typically measured quantitatively by a medical device and then analyzed by a physician or specialist to assess a patient's current status. LLMs are capable of processing these encoded time series signals. For instance, they can decode EEG signals to generate sentences that approximate the semantic content of an individual's thoughts [85] and automatically generate high-quality cardiac diagnosis reports from ECG signals as well as perform zero-shot cardiovascular disease detection [86]. Furthermore, guided by natural language prompts, LLMs can integrate time series signals with rich, patient-specific contextual information to perform critical tasks such as anomaly detection (e.g., arrhythmias), semantic segmentation (e.g., heartbeat or breath phases), and boundary detection, which can lead to earlier diagnoses and more personalized patient care [87].

3.4. Challenges in multimodal medical data

Data heterogeneity represents a fundamental challenge in multimodal medical analysis, encompassing fundamental incompatibilities across multiple dimensions, including file formats, data structures, and semantic interpretation [88]. This heterogeneity is primarily manifested in the significant differences in data formats, structures, and representations. Medical imaging adheres to the DICOM standard, EHR data is commonly structured in HL7 or FHIR formats, and

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clinical notes exist as free text. Meanwhile, physiological signals, such as EEG and ECG, are typically represented as binary waveform data with specific sampling rates. Creating a unified representation for these disparate data types that can be uniformly processed is in itself a highly complex task [89]. Furthermore, semantic and temporal heterogeneity introduces a deeper layer of complexity to data integration. Data from different modalities are typically collected at various time points and for distinct clinical purposes, exhibiting disparate levels of interpretability. A coherent clinical narrative can only be constructed by interpreting these data in the correct temporal sequence and an appropriate manner; failure to do so may adversely affect the model's final decisions [90].

Data security serves as the cornerstone for maintaining patient and institutional trust. In a multimodal context, the risk of data re-identification is significantly amplified. Even after data has been anonymized (e.g., by removing direct identifiers such as names and medical record numbers), the possibility of re-identifying a specific individual persists when information from multiple dimensions is combined [89]. During their training process, LLMs may memorize unique combinations of this information and inadvertently disclose such sensitive data in subsequent interactions, leading to catastrophic privacy breaches [91]. While stringent privacy and security regulations have been enacted to protect privacy, they severely restrict the use of data in the medical field, thereby impeding the acquisition of data required for model training [4]. Regulations such as the Health Insurance Portability and Accountability Act (HIPAA) in the United States and the General Data Protection Regulation (GDPR) in the European Union impose strict restrictions on the processing, storage, and use of personal health information. The acquisition of data for model training often necessitates complex and protracted ethical reviews and institutional approval processes, rendering large-scale, high-quality datasets suitable for training LLMs exceedingly scarce [92].

In addition to being difficult to access, available medical data itself often faces the dual challenges of insufficient quantity and low quality. The issues of data scarcity and imbalance are particularly prominent. For many diseases, especially rare ones, the number of available cases for collection is inherently limited. Even for common diseases, there is often a significant lack of data for specific populations, such as ethnic minorities or certain age groups [88]. A model trained on a data-rich population often shows a sharp decline in performance when applied to a data-scarce population. Concurrently, data quality issues are widespread and manifest in various forms. Incompleteness, such as the absence of a data modality or missing values for certain key fields within a patient's record, is the norm rather than the exception [92]. Inconsistencies, such as significant variations in how clinicians document the same symptom in clinical notes or human errors during data entry, introduce noise into the model's learning process [91]. A more fundamental issue arises from the deep-rooted biases embedded within the data. As reflections of the real world, medical datasets inevitably encode various societal and historical biases, including those related to race, gender, and socioeconomic status [4].

Within the supervised learning paradigm, a critical step for model training is the preparation of data with ground-truth labels. In the medical field, this creates a well-known labeling bottleneck, as producing high-quality annotations for vast amounts of healthcare data is an exceptionally complex, expensive, and expertise-intensive task [91]. The difficulty stems primarily from its heavy reliance on domain-specific expertise, requiring that the annotations be performed by highly trained specialists whose time is exceptionally costly. Furthermore, the complexity of annotation is substantially magnified in a multimodal context. Annotation is no longer confined to a single modality but demands the establishment of precise cross-modal correspondences. For instance, an ideal annotation task might require an annotator to delineate a specific lesion on an MRI image, concurrently identify the sentence describing it in the corresponding clinical notes, and link this to the paragraph in the pathology report that confirms its nature. This degree of multimodal alignment and annotation necessitates collaboration among experts from various domains, making the coordination costs and implementation challenges practically prohibitive [93].

4. Key challenges and solutions

The implementation of Multimodal RAG in the medical domain is confronted with significant technical hurdles. This section systematically analyzes the core challenges, covering semantic alignment, data quality, trustworthiness, complex reasoning, and scalability, and discusses their corresponding representative solutions.

4.1. The semantic gap and alignment problem between modalities

A core and fundamental challenge in Multimodal RAG systems stems from the inherent differences between data modalities, known as the semantic gap. The structure, statistical properties, and information encoding methods of data—such as image pixels, text tokens, and audio waveforms—are fundamentally distinct. Consequently, transforming this heterogeneous data into a unified and comparable representation has become a primary challenge for enabling effective cross-modal retrieval [94]. Early approaches often relied on converting all non-textual modalities into textual descriptions before applying a text-only RAG process. While straightforward, this method inevitably results in significant information loss, as textual descriptions struggle to fully capture the rich details and complex semantics of the original image [35]. Furthermore, even when all modalities are mapped into a unified embedding space, this can give rise to the modality gap problem, where the model exhibits a bias toward retrieving items of the same modality as the query, thereby hindering the accuracy of cross-modal retrieval [95]. In practical multimodal medical RAG systems, these representation issues directly manifest as critical

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retrieval failure modes. For instance, due to weak image-text alignment and same-modality bias, a system queried with a patient's X-ray might retrieve visually similar but clinically irrelevant images, while entirely failing to retrieve highly pertinent textual diagnostic guidelines. This failure to effectively bridge the semantic gap prevents the system from synthesizing comprehensive cross-modal evidence, ultimately compromising the reliability of clinical decision-making.

As a foundational step to bridge the basic semantic gap, the mainstream approach is to train models like VLMs using contrastive learning, most notably CLIP [13]. The core idea of these models is to construct a shared embedding space. They typically consist of an image encoder and a text encoder and are trained on a large number of (image, text) pairs. The learning objective is to maximize the cosine similarity of matching image-text pairs in the embedding space while minimizing the similarity of non-matching pairs. By pulling positive samples closer and pushing negative samples apart, the model learns to map cross-modal content with similar semantics to adjacent positions in the vector space [28]. This technique has been widely applied in practical Multimodal RAG systems. For instance, the SeBe-VQA framework enhances the performance of VQA tasks by extending a pre-trained MLLM (LLaVa-1.5) as a multimodal feature encoder and utilizing a contrastive loss function to align multimodal queries with their corresponding knowledge [96].

Although these pre-trained models establish a shared space, they often still suffer from the aforementioned same-modality bias and struggle with precise alignment in highly specialized domains like healthcare. Therefore, task-specific fine-tuning is a crucial step to explicitly overcome the modality gap. The MMed-RAG framework provides a paradigm for addressing the alignment problem in medical vision-language models. This method fine-tunes the model by constructing a preference dataset that includes positive samples (e.g., correct medical images matching the query) and negative samples (e.g., noisy or irrelevant images for the same query). By training with techniques such as direct preference optimization (DPO), the model learns to prioritize more accurate and relevant multimodal inputs, thereby significantly improving its cross-modal alignment capabilities [22].

4.2. Challenges in data quality and completeness

The performance ceiling of Multimodal RAG systems is limited by the quality of the knowledge base they rely on. This represents a classic "garbage in, garbage out" problem. If the retrieved documents, images, or any other modality of data inherently contain erroneous information, contradictory viewpoints, or are irrelevant to the user's query, the downstream generative model may not only fail to discern their veracity but could also amplify these flaws, ultimately producing content that appears plausible yet is factually incorrect or misleading [97]. This represents a significant challenge, as real-world data is inherently noisy and imperfect.

When faced with the potentially unreliable quality of a knowledge base, a pragmatic approach is to enhance the model's own ability to resist and process noisy data. The RagVL framework proposed by Chen et al. [98] adopts a strategy of actively injecting noise into the training data. During the training process, it intentionally introduces visual noise, for instance, by using corrupted images or applying perturbations at the token level. This proactive noise-injection strategy compels the model to distinguish between signal and noise during the learning process, thereby enhancing its robustness when confronted with noisy real-world data. The RAAT framework, introduced by Fang et al. [99], provides a more nuanced classification of retrieval noise, such as "relevant but without the answer", "counterfactual", and "completely irrelevant". It then utilizes an adaptive adversarial training method to dynamically adjust the model's training process in response to different types of noise. This enables the model to learn targeted coping strategies, rather than just generically resisting all noise.

Beyond data quality, the scarcity and imbalance of data also present a major bottleneck to the advancement of Multimodal RAG technology. For specialized domains such as finance, medicine, and law, acquiring precisely annotated multimodal data is especially difficult, which limits the application of Multimodal RAG in these high-value areas. Annotation tasks that require complex reasoning, for example, creating question-answer pairs for multimodal content like images and text in documents, have extremely high labor costs, leading to limited scales for relevant benchmark datasets [100]. In real-world knowledge bases, data can be incomplete; for instance, an image may lack an accompanying text caption, or a video may be missing an audio track. How to effectively handle this modally incomplete data is a practical challenge that Multimodal RAG systems must address in real-world deployments [101].

Constructing large-scale, high-quality, and modally balanced multimodal datasets is a costly and time-consuming endeavor [102]. Faced with the challenge of scarce training and evaluation data, leveraging powerful generative models to create data has emerged as an efficient and cutting-edge solution. Given a document corpus, an LLM can be utilized to automatically generate natural language questions related to the document content. In this way, large-scale (query, document) pairs can be created for unlabeled documents to train retrievers [103]. The SMMQG framework proposed by Wu et al. [100] demonstrates more complex synthesis capabilities. Through a collaborative workflow that integrates a retriever, an LLM, and an MLLM, it can automatically generate complete multimodal question-answer pairs—including the question, answer, and corresponding source attributions—from multimodal documents like Wikipedia. Research shows that the quality of datasets generated in this manner is comparable to that of human-crowdsourced benchmarks. For the problem of incomplete modalities in datasets, the RAGPT framework proposed by Lang et al. [101] offers an innovative, retrieval-based solution. When the model encounters a target instance with

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a missing modality (e.g., an image), the system searches the knowledge base for other, modally complete instances that are most similar based on the existing modality (e.g., text). It then utilizes the corresponding modality from these complete instances (e.g., their images) via a missing modality generator to reconstruct or impute the target instance's missing modality. Finally, this imputed data, along with dynamic prompts derived from the retrieved content, is fed into the main model for processing. This represents an innovative paradigm that extends the application of RAG from answering user queries to data repair and imputation, providing a powerful tool for handling imperfect real-world data.

4.3. The need for trustworthiness: explainability and reliability

As Multimodal RAG systems become increasingly complex in their architecture and capabilities, their internal decision-making processes can become more opaque. Compared to using only generative models, Multimodal RAG introduces unique potential failure modes, such as the retriever selecting irrelevant entries from the knowledge base or the generator producing misinterpretations despite receiving the correct context [104]. In high-stakes application domains like finance, law, and healthcare, merely providing a correct answer is far from sufficient. To gain user trust and adoption, systems must be reliable and fair, and the reasoning process through which they reach conclusions must be transparent and explainable. The trustworthiness of RAG systems is a multidimensional concept, encompassing aspects such as factuality, robustness, fairness, privacy protection, explainability, and accountability [105]. How to enhance the trustworthiness of Multimodal RAG systems across these multiple dimensions represents a significant direction for future research.

A fundamental requirement of explainability is the ability to trace the model's output back to its original sources. Multimodal RAG can support this by providing the retrieved content on which the answer is based alongside the generated text. To advance this capability, MCiteBench was developed as the first benchmark specifically for evaluating the ability of MLLMs to generate text with accurate citations in multimodal contexts [106]. Furthermore, beyond providing accurate citations, a trustworthy medical system requires robust confidence calibration. It must accurately express its own uncertainty (e.g., by outputting explicit probability scores or using clinical hedging phrases such as "suggestive of, but not definitive for") when the retrieved medical evidence is conflicting, ambiguous, or insufficient, thereby reducing the risk of overconfident but incorrect diagnoses. For more complex reasoning processes, explainability requires the system to display its intermediate reasoning steps. This necessitates the development of techniques that can trace the flow of information through multi-hop retrieval paths—for example, technologies that can clearly demonstrate the roles and impacts of different nodes and edges in a knowledge graph during the reasoning process [107]. Furthermore, a challenging research direction is to develop techniques that can quantify and explain the contribution of different modalities to the final output. For instance, the system would need to answer questions like, "Which contributed more to the final answer: the image or the text in the Multimodal RAG context?" thereby providing users with deeper insights into the decision-making process.

To systematically evaluate and enhance the reliability of Multimodal RAG, researchers have developed a series of specialized metrics and frameworks. The RAG-Check framework proposed by Mortaheb et al. [108] ingeniously decouples the reliability assessment of Multimodal RAG into two independent parts. It trains two specialized models: an RS model to evaluate the relevance between the retrieved multimodal data and the user query, and a CS model to assess whether the system-generated answer is factually consistent with the retrieved context. This decoupled evaluation approach can precisely diagnose whether the system's performance bottleneck lies in the retrieval or the generation stage. RULE, proposed by Xia et al. [109], is a reliability enhancement framework designed specifically for medical vision-language models. It enhances factuality through two core components: first, it employs a calibrated context selection strategy, managing factuality risks by controlling the number of retrieved contexts; second, it utilizes a carefully constructed preference dataset to fine-tune the model to address the issue of over-reliance on retrieved information. This teaches the model not to be swayed by irrelevant or incorrect retrieved content when its own internal knowledge is already correct.

4.4. Domain adaptation and generalization

Despite possessing extensive general knowledge, pretrained large models often encounter performance bottlenecks when applied to specialized domains such as healthcare, finance, law, or specific industries. This phenomenon, known as "domain shift", occurs when the data distribution on which the model was trained does not align with the data distribution of a specific domain. General-purpose models may fail to comprehend the specialized terminology, abbreviations, and implicit knowledge inherent to a particular domain.

Domain-specific fine-tuning is the most direct and effective method for adapting general-purpose models to a specific domain. Its core idea is to further train a pretrained model using a smaller, high-quality, domain-specific dataset. Through this process, the model learns the linguistic style, specialized terminology, and knowledge patterns of the specific domain, thereby significantly improving its performance in that area. The primary bottleneck for fine-tuning is the acquisition of high-quality, domain-annotated data. To address this issue, automated fine-tuning frameworks have emerged. For example, the ALoFTRAG framework demonstrates a self-supervised approach: it utilizes a locally deployed LLM to first automatically generate question-answer pairs and hard negatives from unlabeled domain documents and then uses this generated data to fine-tune itself [110].

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This method requires no manual annotation; as the entire process is executed locally, it effectively safeguards the privacy of sensitive data in fields like finance and healthcare, offering a viable path for low-cost, high-efficiency domain adaptation.

While standard fine-tuning can inject domain knowledge, it does not explicitly train the model on how to utilize retrieved documents to answer questions. RAFT, proposed by Zhang et al. [111], is an innovative training method specifically designed for "open-book exam" style RAG scenarios. During the RAFT training process, the model is presented with a question, the golden documents from which the answer can be derived, and several irrelevant distractor documents. The model's task is to learn to ignore the distracting information and generate a well-reasoned, cited answer based on the content within the golden documents. This training approach not only enables the model to learn domain knowledge but, more importantly, teaches it how to perform effective information filtering and utilization amidst noisy retrieval results, thereby significantly enhancing its robustness in real-world RAG applications.

In complex environments with multiple RAG applications, deploying a dedicated retriever for each application is often cost-prohibitive. To address this challenge, a more scalable strategy is to train a single, general-purpose retriever that can serve multiple domains. The multi-task retriever fine-tuning proposed by Béchard and Ayala [112] is designed precisely for this purpose. This method involves training a small, instruction fine-tuned retriever on a combined dataset of various domain-specific tasks, such as simultaneously learning to retrieve workflow steps, database table names, and field names. This approach not only enhances the model's retrieval accuracy within a specific domain but also enables it to generalize effectively to out-of-domain settings and new retrieval tasks. Ultimately, a single lightweight retriever can efficiently support multiple RAG applications across the ecosystem, achieving low costs and high scalability while reducing the coupling between the retriever and downstream LLMs.

4.5. Knowledge synthesis and complex reasoning

As Multimodal RAG systems evolve, the challenges they face are progressively shifting from simple information retrieval to the demand for complex, multi-step reasoning capabilities. Many real-world problems cannot be answered with information from a single retrieval, requiring the system to possess the ability for multiple retrievals and multi-hop reasoning [113]. The system must synthesize and chain information from various documents or different modalities, deriving conclusions step-by-step to reach the correct answer. The traditional RAG pipeline, with its typical single-retrieval and fixed-length chunking strategy, proves inadequate for such complex reasoning tasks. This design disrupts the global contextual structure of documents and fails to provide the model with the complete evidence chain required for complex reasoning [114].

To address the challenges of complex reasoning, one research direction is to reconstruct Multimodal RAG from a static pipeline into a dynamic, agent-driven process. This approach endows the system with greater flexibility and autonomy. In this method, an agentic LLM acting as a planner first decomposes a complex, high-level user query into a series of simpler, directly executable sub-problems [115]. For example, the HM-RAG framework proposed by Liu et al. [116] includes a specialized Decomposition Agent to perform the user query decomposition task. The OmniSearch system adopts a similar idea, capable of dynamically decomposing complex multimodal questions into a sub-question chain that includes specific retrieval actions [117]. Furthermore, unlike single retrieval, Agentic RAG should be able to execute multi-round retrieve-generate cycles. The system first conducts an initial retrieval, then uses the retrieved information to generate a preliminary answer or identify current knowledge gaps. Subsequently, it uses this intermediate result as new context to formulate a more targeted second-round retrieval query, repeating this process until sufficient information is gathered to answer the original question [113]. This synergy between retrieval and generation greatly enhances the model's ability to collect a complete evidence chain.

The Chain-of-Thought (CoT) prompting technique, which stimulates step-by-step thinking in models, can also be applied to multimodal scenarios. The model can produce intermediate thought processes and reasoning steps based on multimodal information before generating the final answer. In Multimodal Chain-of-Thought (MCoT), a key challenge is providing the model with the most effective and relevant demonstrations. The work proposed by Liu et al. [118] addresses this by dynamically selecting optimal CoT demonstrations through retrieval and stratified sampling techniques, thereby significantly improving the model's reasoning performance. For tasks requiring reasoning over structured relationships between entities, integrating knowledge graphs is a powerful solution. The research by Lee et al. [119] introduces Multimodal Knowledge Graphs (MMKGs), leveraging their rich cross-modal semantic knowledge to empower large language models, thus significantly enhancing their comprehensive multimodal reasoning capabilities.

Furthermore, another approach is reasoning-enhanced reranking based on reinforcement learning (RL). This involves using RL to train a reranker that not only assesses document relevance but also understands and prefers documents that support a coherent reasoning path. The MM-R5 framework proposed by Xu et al. [120] is representative of this direction. Its training is divided into two stages: first, through supervised fine-tuning (SFT), the model learns to generate reasoning chains from high-quality, human-annotated reasoning data; then, in the second stage, it is optimized using reinforcement learning. The RL reward function is meticulously designed to consider not only the accuracy of the reranking but also to include an evaluation of the quality of the generated reasoning chain. Consequently, the model

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is explicitly trained to prioritize documents that can form a logically clear and complete reasoning chain.

4.6. Scalability and deployment challenges

As the scale of knowledge bases expands from thousands to millions or even billions of entries, the scalability of Multimodal RAG systems presents a significant challenge. The computational cost and latency of performing similarity searches become major bottlenecks constraining their real-world application. Traditional exact nearest neighbor search algorithms, whose time complexity scales linearly with the dataset size, result in unacceptable response times for large-scale knowledge bases, rendering them infeasible for real-time applications. Non-textual data such as images and videos are typically encoded into high-dimensional feature vectors, which inherently increases computational complexity. Furthermore, the large volume of media files poses challenges for storage and processing. The resource overhead for large-scale computation, along with costs associated with index construction, retrieval latency, and maintenance, collectively contribute to the scalability dilemma faced by Multimodal RAG systems.

To achieve large-scale, low-latency retrieval while maintaining acceptable accuracy, Multimodal RAG systems can employ approximate nearest neighbor (ANN) search algorithms instead of pursuing exact matches. Among the various ANN algorithms, graph-based methods, particularly Hierarchical Navigable Small World (HNSW), have become one of the most effective techniques in the industry. The core idea of HNSW is to organize data points (i.e., vectors) by constructing a multi-layered graph structure. Its coarse-to-fine search strategy enables the achievement of logarithmic time complexity, thereby providing exceptional scalability [121].

In addition to underlying ANN indexing algorithms, optimizing macro-level retrieval strategies is crucial for enhancing system efficiency. For instance, coarse-to-fine and hierarchical retrieval strategies can avoid performing high-precision searches across the entire knowledge base for every query. The OMGM system proposed by Yang et al. [122] first performs a rapid preliminary screening at the coarse-grained level of document summaries and only then invokes a more computationally expensive multimodal reranking model on a small set of top-k candidates for refined ranking. Similarly, MultiFinRAG, introduced by Gondhalekar et al. [123], employs a tiered fallback strategy. The system initially retrieves only lower-cost textual information and only falls back to retrieving higher-cost image and table data when the textual information is insufficient to answer the query. Finally, generative retrieval offers a paradigm-shifting alternative to traditional embedding-based retrieval. Instead of calculating vector similarities to find matches, generative retrieval trains a model to directly generate the unique identifier (ID) of a target document from a user query. The retrieval time of this method is independent of the knowledge base size, achieving constant-time complexity and fundamentally addressing the scalability problem. The GENIUS model proposed by Kim et al. [124] is a representative work in this direction.

In practical production deployments, certain components of a Multimodal RAG system can adopt lightweight models, such as those that are quantized or have a smaller parameter count, provided that acceptable performance is maintained. This approach can significantly reduce computational load and memory footprint, thereby accelerating overall response times and making it feasible to deploy complex Multimodal RAG systems on commodity hardware. Furthermore, optimizing the system architecture is essential. Employing asynchronous processing mechanisms allows for the parallel execution of multiple retrieval tasks, preventing process bottlenecks. Concurrently, sharding the vast index and deploying it across multiple compute nodes to form a distributed system is an effective means of handling massive datasets and high-concurrency requests.

5. Applications of Multimodal RAG in medicine

Multimodal RAG is evolving from theoretical frameworks to promising clinical applications. This section surveys its diverse use cases within the medical domain, categorized into five key areas: diagnostic assistance, medical question answering, automated report generation, education, and personalized treatment. We illustrate how leveraging external multimodal knowledge enhances the accuracy and efficiency of medical services across these scenarios.

5.1. Clinical diagnostic assistance

Clinical diagnostic assistance is one of the most impactful application areas for Multimodal RAG technology. The application of Multimodal RAG in diagnostic support covers multiple stages, from preliminary screening to complex differential diagnosis. By integrating multimodal patient data and leveraging external knowledge bases for fact-checking, it can provide clinicians with more accurate and comprehensive diagnostic suggestions, effectively reducing the risk of misdiagnosis.

Differential Diagnosis: The system can generate a probability-ranked list of differential diagnoses based on a patient's medical history, physical signs, and symptom descriptions, combined with knowledge retrieved from clinical guidelines or medical literature. For example, a RAG-enhanced GPT-4 model increased its accuracy in identifying the main diagnosis from 54% to 78% [125].

Image-based Diagnosis: In the fields of radiology and pathology, Multimodal RAG systems can directly analyze medical images and provide diagnostic support. RAG can integrate image retrieval-augmented systems; for example, by retrieving historical cases and their confirmed diagnoses similar to the current ultrasound image to assist in breast cancer classification [126], or by retrieving from relevant knowledge bases to aid in the diagnosis of pathological slides [22].

EHR Interpretation: Clinical records contain a vast amount of unstructured text information. A RAG system

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can perform in-depth interpretation of this text while ensuring patient privacy. It can answer a clinician's queries about specific diagnoses, for example, "Based on this cardiac catheterization report, does the patient have a paravalvular leak (PVL)?" [127].

5.2. Medical question answering

Medical question answering is another core application of Multimodal RAG, aimed at enabling clinicians, researchers, and others to interact efficiently with vast amounts of multimodal medical data through natural language. Unlike traditional search engines, Multimodal RAG-driven QA systems can provide more direct, contextual, and fact-based answers instead of a long list of document links. Multimodal RAG demonstrates broad applicability in the medical QA domain, showing immense potential in applications ranging from basic knowledge retrieval to complex clinical scenario analysis requiring deep reasoning.

Visual Question Answering: Medical VQA is one of the fastest-growing and most challenging sub-fields, revolutionizing how humans interact with medical images. It allows clinicians, researchers, and even patients to ask questions about medical images directly in natural language, thereby greatly enhancing the interactivity and efficiency of image analysis [11]. For instance, in a Multimodal RAG application, a clinician can inquire about the presence of an abnormality in an image. The system then retrieves relevant information using the image and the question, constructs a retrieval prompt, and thereby guides the model to generate a more accurate answer [128].

Clinical Guideline and Literature Search: In daily clinical practice, physicians need to quickly access the latest diagnostic and treatment information. A Multimodal RAG system can act as an intelligent medical knowledge base assistant. For complex clinical questions posed by users, the system can retrieve and synthesize information from the latest clinical guidelines, authoritative medical textbooks, or PubMed literature to provide answers grounded in clear evidence.

EHRs and Complex Documents Interaction: A patient's EHR contains a wealth of unstructured text, scanned PDF reports, and embedded charts, all of which are critical for understanding the patient's condition. Physicians can query a patient's unstructured EHR within a fully localized, HIPAA-compliant RAG system, where the system not only provides an answer but also offers citations to specific passages in the original documents, ensuring the traceability of the answer [127]. For complex documents containing intricate charts and data, vision-guided document chunking technology can ensure that the system correctly understands the correspondence between text and graphics to answer user queries accurately [129].

5.3. Automated medical report generation

Automated medical report generation, particularly for imaging-intensive disciplines such as radiology and pathology, is one of the most practical applications for Multimodal RAG technology. Its core objective is to automatically generate a diagnostic report that is factually accurate, linguistically fluent, and structurally compliant with clinical standards, based on input medical images and other relevant patient information. This task is inherently a complex multimodal-to-text conversion process, highly susceptible to factual hallucinations where the generated text does not align with the visual content. The Multimodal RAG framework can address this by retrieving similar cases from a large, high-quality database of image-report pairs to serve as a reference before generation, providing a solid factual basis for the generation process. This significantly improves the accuracy, completeness, and reliability of the generated reports, effectively alleviating the documentation burden on physicians and enhancing the efficiency of clinical workflows.

Radiology Report Automation: This is currently the most intensively researched and rapidly developing area, aimed at addressing the high workload and repetitive labor faced by radiologists. Multimodal RAG systems can significantly enhance the automation of report writing. For example, a system can automatically analyze a chest X-ray and generate a standardized report containing sections like "Findings" and "Impression" [11].

Pathology Report Generation: With the rise of digital pathology, AI has also shown immense potential in pathology report generation. A multimodal vision-language model can analyze digitized whole slide images (WSIs), identify key pathological features such as cell morphology and tissue architecture, and generate descriptive pathology reports [130]. Pathologists often refer to similar past cases during diagnosis. By integrating Multimodal RAG technology, this diagnostic workflow can be simulated: retrieving WSIs with similar visual features and their confirmed reports from a database provides a strong reference for generating the report for the current case, thereby ensuring diagnostic consistency and accuracy [131].

General-Purpose Frameworks for Cross-Specialty Report Generation: To accommodate the needs of different medical specialties, researchers are working to build more versatile Multimodal RAG frameworks. General-purpose frameworks like MMed-RAG are designed to handle a variety of medical images, including those from radiology, ophthalmology, and pathology. Through its "domain-aware" retrieval mechanism, the system can match the most appropriate knowledge sources to different types of images, thereby achieving significant improvements in factual accuracy across report generation tasks in multiple specialties [22].

5.4. Medical education and training

Multimodal RAG technology shows promising potential in medical education by facilitating the creation of highly

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realistic, interactive simulation environments that provide medical students and junior doctors with opportunities to practice clinical skills in a safe setting. Whereas traditional medical education relies on textbooks, lectures, and limited standardized patient interactions, Multimodal RAG offers an emerging approach to immersive and personalized learning by integrating multiple data modalities (e.g., video, audio, and text) and external knowledge bases. The system aims to simulate dynamic and unpredictable clinical scenarios, potentially transforming students from passive knowledge recipients into active participants, thereby helping to cultivate clinical reasoning, communication skills, and decision-making abilities.

Interactive Virtual Patients: By combining physical social robots with rich facial expressions with powerful LLMs, researchers have created highly realistic virtual patients. The system can simulate patients with specific medical histories, and medical students can interact with these robots through natural language to conduct consultations and collect medical histories [132]. By incorporating Multimodal RAG technology, the system can leverage information retrieved from knowledge bases (such as past medical records) to further simulate patients with diverse personalities, medical histories, and communication styles, while ensuring the factual accuracy of their responses. This embodied interaction greatly enhances the realism of the simulation and the students' emotional engagement, enabling them to practice clinical reasoning and communication skills in a hands-on manner.

Surgical Skill Analysis: For hands-on disciplines like surgery, Multimodal RAG shows immense potential for providing fine-grained, objective feedback. For example, a system named Multi-RAG is designed to understand complex multimodal information streams containing video, audio, and text documents [133]. In surgical training, such a Multimodal RAG system can be used to analyze surgical recordings (video modality) in conjunction with standard operating procedures (text modality) and intraoperative dialogue (audio modality). Trainees can ask the system questions about the surgical content, and by retrieving relevant information, the system can provide evidence-based, quantifiable feedback crucial for skill improvement.

Personalized Learning Platforms: Large multimodal medical datasets (such as MedPix®) are themselves valuable educational resources, containing numerous annotated images and case descriptions [134]. Multimodal RAG systems can leverage these datasets to provide an interactive knowledge base for physicians, nurses, and medical students to query and learn from. The system can also automatically retrieve and recommend the most relevant medical materials based on user interactions, enabling personalized learning paths and helping users efficiently address knowledge gaps.

5.5. Personalized treatment planning

Personalized treatment, the cornerstone of precision medicine, aims to customize the most effective and safest therapeutic regimens based on each patient's unique biological characteristics (e.g., genomics), clinical manifestations (such as imaging features and EHRs), and individual profile. The application of Multimodal RAG technology in this domain represents its most sophisticated use case, requiring systems capable of integrating highly diverse and complex data sources. These systems must perform deep, evidence-based reasoning to assist clinicians in making optimal individualized decisions.

Genomics-Guided Precision Therapy: This is one of the most cutting-edge and data-intensive applications of personalized medicine. By integrating a patient's genomic sequencing data, EHR, pharmacological information, and the latest clinical research, Multimodal RAG systems can provide precise therapeutic recommendations for complex diseases like cancer. A feasible approach involves integrating Multimodal RAG with GraphRAG technology. This leverages knowledge graphs to connect a vast network of information, including genetic variations, related proteins, targeted drugs and their mechanisms of action, and clinical guidelines. When presented with a cancer patient carrying a specific gene mutation, the system can perform complex multi-hop reasoning to retrieve targeted drugs that act on the relevant pathway. Combined with the patient's EHR information and the latest clinical guidelines, it can ultimately recommend the most suitable personalized medication regimen and provide a clear reasoning path.

Personalized Disease Prediction and Health Management: The application of Multimodal RAG is extending from treatment to prevention. Frameworks such as Health-LLM can perform in-depth analysis of user health reports, which contain both structured data and textual descriptions. By employing a RAG mechanism to retrieve relevant information from extensive medical knowledge bases, these systems can predict an individual's future risk of developing specific diseases, such as diabetes and heart disease [135]. Based on these personalized risk predictions, the system can further provide specific, actionable recommendations for lifestyle modifications, dietary guidance, and health screenings, helping users engage in proactive and forward-looking health management.

Automated Multimodal Surgical Planning: In the surgical domain, Multimodal RAG systems can assist clinicians in formulating highly detailed and personalized surgical plans. A prime example is the FUAS-Agents system, an autonomous multi-agent framework designed for Focused Ultrasound Ablation Surgery (FUAS) [136]. This system receives a patient's profile and MRI scans, then autonomously orchestrates a suite of specialized AI tools to execute subtasks. Concurrently, it retrieves and integrates relevant clinical surgical guidelines via RAG, ultimately generating a comprehensive, personalized treatment plan that incorporates both graphics and text.

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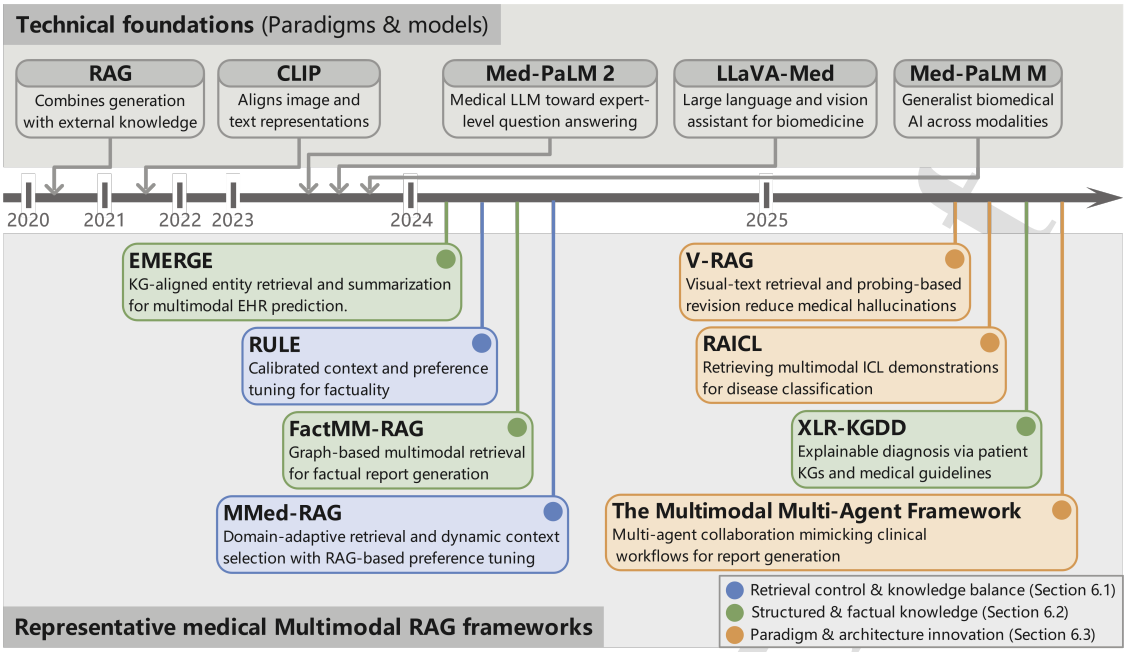


Figure 3: Chronological evolution of medical Multimodal RAG. The upper timeline highlights foundational paradigms and models (discussed in Section 2). The lower section categorizes representative frameworks (discussed in Section 6) into three dimensions: retrieval control and knowledge balance (blue), structured and factual knowledge (green), and paradigm and architecture innovation (orange).

6. Overview of representative works

In this section, we provide a comprehensive review of representative frameworks in the field of medical Multimodal RAG. Unlike general domain approaches, medical scenarios impose stricter requirements on factuality, reasoning capability, and the handling of heterogeneous data. To systematically analyze these advancements, we categorize the existing works into three distinct dimensions: (1) retrieval control and knowledge balance, which focuses on optimizing the interaction between internal parameters and external knowledge; (2) integration of structured and factual knowledge, which emphasizes shifting from simple semantic similarity to verified evidence for higher clinical reliability; and (3) innovation in paradigms and architectures, which explores novel workflow designs and reasoning mechanisms beyond traditional retrieval-generation pipelines. To visualize the developmental trajectory and categorization of these representative works, we present a comprehensive timeline in Figure 3.

6.1. Frameworks for retrieval control and knowledge balance

The research in this section is dedicated to addressing the fundamental challenges faced when applying RAG to Medical Large Vision-Language Models (Med-LVLMs). These challenges include how to precisely control the retrieval process and how to mitigate the risks associated with the model's over-reliance on external knowledge, which may contain noise.

RULE: Xia et al. [109] were among the first to systematically identify two key failure modes when applying RAG in medical scenarios: first, retrieved contexts can be either insufficient (lacking information) or excessive (introducing noise and interference); second, the model may erroneously trust the retrieved information, even when its own parameterized knowledge is initially correct. To address these issues, they proposed the RULE framework. For the problem of an arbitrary choice of the retrieval number, k, which may introduce irrelevant or erroneous information, RULE introduces a provably effective strategy for controlling factuality risk. This strategy calibrates the value of k through hypothesis testing on a separate validation set, theoretically ensuring that the factuality risk is controlled within an acceptable threshold. However, the validation set it relies on may not always be available or easy to construct in practical applications. Second, to tackle the problem of models over-relying on retrieved content, which can lead to errors, RULE proposes Knowledge Balanced Preference Tuning (KBPT). This method begins by curating a specific preference dataset from samples where errors arise from an over-reliance on retrieved contexts. Subsequently, DPO is used to fine-tune the model on this dataset, explicitly teaching it to better balance its dependence on its internal parameterized knowledge and the external retrieved knowledge. However, its KBPT strategy only addresses a specific failure scenario—"correct without RAG, incorrect with RAG"—and does not explicitly resolve deeper alignment issues, such as cross-modality misalignment where the model overly relies on retrieved

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text while insufficiently grounding its reasoning in the input medical image.

MMed-RAG: Subsequently, the MMed-RAG framework proposed by Xia et al. [22] can be seen as an evolution of the RULE framework, featuring enhanced versatility and robustness. Medical images are inherently heterogeneous, covering multiple domains such as radiology, pathology, and ophthalmology, each with unique visual features and textual descriptions. Consequently, traditional RAG methods that rely on a single retriever exhibit limited generalizability and suboptimal performance. Furthermore, the direct injection of external knowledge can potentially disrupt the model's original internal alignment, leading to cross-modal misalignment (i.e., over-reliance on retrieved text while neglecting the visual input) and overall misalignment with the ground truth (i.e., generating incorrect responses due to misleading retrieved information). The MMed-RAG framework introduces a domain identification module that first classifies the domain of the input image and then selects a domain-specific retriever for knowledge retrieval. It then adaptively determines the number of contexts by dynamically evaluating the similarity scores of the retrieved results, effectively preventing low-quality or irrelevant documents from interfering with the generation process. The framework also proposes a more sophisticated RAG-based preference fine-tuning (RAG-PT) strategy, which uses carefully curated cross-modality and overall alignment preference data to improve both types of alignment. This fine-tuning process involves three types of samples, designed to encourage the model to prioritize the input medical image when generating responses and to learn how to appropriately handle retrieved information in various scenarios. By establishing a more comprehensive strategy that integrates retrieval, context selection, and generation alignment, MMed-RAG lays the foundation for building more robust and versatile medical Multimodal RAG systems. However, its powerful capabilities come at the cost of significantly increased complexity. Constructing such a refined preference dataset for RAG-PT is a data-intensive and procedurally complex process, and the deployment of the framework also imposes higher demands on storage and computational resources.

6.2. Frameworks based on structured and factual knowledge

In this section, we analyze Multimodal RAG frameworks in the medical domain that focus on leveraging structured and factual knowledge. These studies represent a critical evolution in retrieval strategies, shifting from simple textual semantic similarity to retrieval grounded in verified facts and structured knowledge, thereby constructing more reliable and explainable systems.

FactMM-RAG: For tasks such as radiology report generation in the medical domain, semantic similarity is far from sufficient. A retrieved report might use similar vocabulary but describe entirely different clinical facts, leading to severe factual inaccuracies in the generated report. The FactMM-RAG framework proposed by Sun et al. [137] introduces an innovative process that utilizes the RadGraph tool to parse free-text radiology reports into structured knowledge graphs composed of clinical entities and their relationships. By comparing these graphs, it then mines factual report pairs, ensuring that retrieved reports share the same underlying clinical observations as the query image, rather than just keywords. The framework subsequently uses these fact-verified report pairs to train the retriever, thereby directly injecting factual medical knowledge and teaching it to align images with reports that are accurate in clinical fact. Consequently, during the retrieval process, the retriever can prioritize the most factually relevant reference report for the query image and propagate this factual advantage to the downstream generation model. By shifting the optimization objective from semantic similarity to factual correctness, it provides a stronger evidence-based signal for the entire RAG pipeline, ultimately significantly improving the clinical accuracy of the generated reports. However, the framework's limitations are also apparent. First, its effectiveness is highly domain-specific, primarily targeting chest radiology. Second, its definition of factuality is entirely dependent on the external RadGraph tool, whose own coverage and accuracy could become a system bottleneck.

EMERGE: Medical data extends far beyond images and unstructured text. For example, EHRs contain vast amounts of structured time-series data, and there is also a massive amount of medical knowledge consolidated in well-curated knowledge graphs. Traditional RAG frameworks struggle to effectively utilize such data. The EMERGE framework, proposed by Zhu et al. [138], focuses on enhancing the predictive modeling capabilities for multimodal EHRs, such as for in-hospital mortality and 30-day readmission prediction. It employs the z-score method and LLMs to extract medical entities from time-series data and clinical notes, aligns them with a professional medical knowledge graph (PrimeKG), and extracts rich information including relationships, definitions, and descriptions. This structured knowledge is then synthesized and distilled into a summary text by a long-context LLM, which is then fused with the original EHR data representations for final prediction via an adaptive multimodal fusion network. This approach greatly enriches the model's input information, allowing data from different sources to cross-reference and supplement each other, generating a more comprehensive final feature vector to understand the patient's condition. However, the disadvantage of the EMERGE framework lies in its extreme procedural complexity. The entire process involves multiple stages, including entity extraction, entity matching, knowledge retrieval, LLM summarization, and multimodal fusion, with each stage posing potential issues. Secondly, the framework's performance is highly dependent on the quality and coverage of the external knowledge graph.

XLR-KGDD: The XLR-KGDD framework, proposed by Bedi et al. [139], concentrates on explainable disease diagnosis. It maps a patient's multimodal clinical information into a structured graph, the patient KG, which serves as a condensed representation of that patient's information.

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Subsequently, this patient KG is used to construct a detailed prompt, which is fed into an LLM for disease diagnosis. To provide a well-founded explanation for the given diagnosis, the framework retrieves relevant paragraphs from standard medical guidelines to serve as context for the LLM. This ensures that the generated explanations are factually consistent with accepted clinical standards, thereby enhancing trust in the AI's diagnostic results. However, the framework's drawback is the need to construct the patient KG, which is a complex and critical prerequisite step. Its diagnostic accuracy depends to a large extent on the quality of the patient KG construction and the effectiveness of prompt engineering, while the RAG component itself does not contribute to improving diagnostic accuracy. Furthermore, the generated explanations may contain factual inaccuracies and ambiguities.

6.3. Frameworks for innovation in paradigms and architectures

The research in this section is dedicated to expanding the boundaries of the traditional RAG paradigm through innovative system architectures and learning strategies. Moving beyond simple information retrieval, these approaches aim to endow systems with stronger reasoning depth, greater adaptability, and better alignment with clinical logic.

V-RAG: Traditional RAG frameworks applied to medical images suffer from a fundamental flaw: they assume that retrieved text reports are a perfect proxy for the retrieved images. This approach overlooks the visual information within the retrieved images themselves, which may contain key nuances that the text fails to capture. The V-RAG framework proposed by Chu et al. [140] aims to mitigate the hallucination problem in Medical MLLMs (Med-MLLMs) during the generation of radiology reports through a multi-stage correction process, thereby enhancing clinical accuracy. Its core idea is that when the model receives a query image, unlike traditional methods that only retrieve and utilize the associated text of similar images, it leverages both the retrieved similar images themselves and their corresponding text reports. The final prompt then contains both the query image and the retrieved image-report pairs. Recognizing that most Med-MLLMs are trained on single images and struggle with multi-image reasoning, the study proposes a series of fine-tuning tasks. These tasks are designed to explicitly teach single-image models how to process multi-image inputs and associate information correctly, thereby making them "V-RAG-capable". In the specific report optimization workflow, the method first utilizes a Med-MLLM to generate an initial report for a given query image and extracts key medical entities from it. Subsequently, it performs entity probing based on similar image-report pairs retrieved from an external database to individually verify the authenticity of these entities. Finally, a powerful LLM is used to rewrite the report based on these more reliable, visually grounded probing results. However, the framework requires specialized fine-tuning for models that can only handle single images. Moreover, the framework's process involves multiple steps—generation, extraction, entity probing, and final rewriting—and requires processing and understanding multiple images during inference, which inevitably increases computational cost.

RAICL: Fine-tuning MLLMs for each specific task or disease is not only computationally expensive but also requires large-scale annotated datasets, which is often infeasible for scenarios such as rare diseases. Zhan et al. [141] proposed a Retrieval-Augmented In-Context Learning (RAICL) framework, a novel paradigm that utilizes RAG to offer a simple, efficient, and scalable alternative to fine-tuning-based approaches. Instead of treating RAG as a tool to inject external knowledge for answering a single query, RAICL leverages RAG to find the most relevant demonstrations for in-context learning (ICL). For a given query, such as a pathology slide, it retrieves the most similar slides and their correct classification results from a database, formatting them as few-shot examples in the prompt to guide the model toward a more accurate prediction. RAICL significantly improves the accuracy of disease classification, demonstrating that RAG can serve as a powerful dynamic prompting tool to adapt MLLMs for few-shot classification tasks without fine-tuning, making it highly valuable for rare disease classification and other low-data scenarios. However, the framework's limitations are that its performance is highly dependent on the powerful ICL capability of the MLLM itself, the choice of the embedding model used for retrieval, and the quality of the retrieved demonstrations.

The Multimodal Multi-Agent Framework: Generating a high-quality radiology report is a complex, multi-step reasoning process that is difficult for single models to replicate perfectly. They often fail to follow the structured, logical workflow of human radiologists. The multimodal multi-agent framework proposed by Yi et al. [142] decomposes the radiology report generation task and assigns it to five specialized agents that mimic the clinical workflow: a Retrieval Agent (to find similar reports), a Draft Agent (to create a preliminary report), a Refiner Agent (to distill key findings), a Vision Agent (to provide an image-based description), and a Synthesis Agent (to integrate all inputs and generate the final report). These agents communicate through structured intermediate outputs, creating an interpretable and controllable reasoning chain. This modular design allows each agent to focus on its specific sub-task, thereby improving overall quality and alignment with clinical practice. This work introduces the multi-agent system (MAS) paradigm to the domain of medical RAG. It shows that for highly complex and structured tasks, decomposing the problem and assigning it to specialized agents can be more effective than training a single, end-to-end model. However, while the modular design offers advantages, it also introduces new risks. Any minor error from an agent early in the pipeline could be amplified and propagated to downstream agents, ultimately affecting the quality of the final report; the combination of retrieved reports, refined findings, and visual descriptions introduces additional complexity, which can affect the overall consistency of the final output.

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Furthermore, the sequential interaction between agents may lead to higher latency than end-to-end models.

7. Datasets and evaluation metrics

This section provides a systematic review of the foundational datasets and evaluation methodologies for medical Multimodal RAG systems. We conduct an in-depth analysis of the medical multimodal datasets that underpin the development of this field, covering a range of resources from large-scale clinical archives to specialized knowledge graphs. Subsequently, we introduce and categorize evaluation metrics according to their assessment objectives. Finally, the performance of representative medical Multimodal RAG frameworks is evaluated and compared, classified by the type of clinical task.

7.1. Overview of commonly used medical multimodal datasets

To facilitate a clear comparison of the foundational resources driving this field, we summarize the main medical multimodal datasets discussed in this survey in Table 1. This table outlines the modality, task, scale, and annotation type of each dataset.

7.1.1. Electronic health records datasets

MIMIC-III [143] is a comprehensive critical care database from Beth Israel Deaconess Medical Center, covering the period from 2001 to 2012. It contains data from 49,785 hospital admissions corresponding to 38,597 distinct adult patients, in addition to records for over 7,800 neonates. Organized as a relational database of 26 tables linked by identifiers such as SUBJECT_ID and HADM_ID, it offers a wide spectrum of clinical data. This includes patient demographics, nurse-verified physiological data like vital signs, laboratory results, medication orders, and fluid balance records. The dataset also contains billing codes such as ICD-9 for diagnoses and DRG codes, as well as de-identified free-text notes, which encompass discharge summaries, radiology reports, and progress notes.

MIMIC-IV [144] represents a major update, encompassing contemporary data from 2008 to 2019 and comprising a total of 431,231 hospital admissions from 180,733 distinct patients. The database is organized using a modular structure, primarily divided into hosp, icu, and note modules that group data by provenance and content. The hosp module contains hospital-wide data such as laboratory results, microbiology cultures, and administrative information derived from billing, including both ICD-9 and ICD-10 diagnosis codes. The icu module provides granular data documented at the ICU bedside, like intravenous infusions and charted observations. The note module contains de-identified free-text discharge summaries and radiology reports. Furthermore, the inclusion of new data sources like the electronic Medicine Administration Record (eMAR) offers precise medication tracking. This rich, multimodal data supports academic research, enabling researchers to undertake a variety of deep learning tasks. These range from classification tasks, such as in-hospital mortality prediction, to advanced NLP applications like clinical report summarization or the generation of reports from structured data.

7.1.2. Radiology datasets

The Indiana University Chest X-ray Collection (IU X-Ray) [145] is the first publicly available collection that pairs radiology images with their corresponding diagnostic reports. It contains 7,470 chest X-ray images (frontal and lateral views) and 3,955 corresponding radiology reports. The reports are clearly structured, containing key sections such as Indications, Findings, and Impression. The data collection was approved by an Institutional Review Board (IRB) and underwent a rigorous de-identification process. IU X-Ray laid a foundational basis for research in fields like medical image retrieval and clinical decision support and played a significant role in pioneering medical report generation research.

MIMIC-CXR [146] is a large-scale, publicly available chest X-ray dataset, comprising 377,110 images from 227,835 imaging studies for 65,379 patients presenting to the Beth Israel Deaconess Medical Center Emergency Department between 2011 and 2016. The dataset provides paired DICOM-format images and free-text radiology reports. The reports are semi-structured free text and, after processing, primarily contain the main body written by a radiologist, which typically includes subsections such as Findings and Impression. MIMIC-CXR is intended to provide a foundational resource for a wide range of research fields, including computer vision, natural language processing, and clinical data mining, and is particularly suitable for developing and validating tasks such as radiology report generation from images and information extraction.

CheXpert [147] is a large dataset designed for automated chest radiograph interpretation, containing 224,316 chest radiographs from 65,240 patients sourced from Stanford Hospital. Its core task is multi-label classification for 14 clinical observations. A key innovation of CheXpert is its automated labeler, which can explicitly extract diagnostic uncertainty from radiology reports, classifying labels as positive, negative, or uncertain. This approach more realistically reflects the complexity of clinical interpretation, moving beyond simple binary judgments. Subsequently, the dataset was expanded into CheXpert Plus [148], which, building upon the original, publicly released paired and extensively de-identified full radiology report texts, patient demographics, DICOM-format images, and RadGraph knowledge graph annotations.

7.1.3. Specialized medical datasets

Oncology datasets: The Cancer Genome Atlas (TCGA) [149] is a landmark cancer genomics program that systematically identifies and discovers major cancer-causing genomic alterations to create a comprehensive genomic map for multiple cancer types. The project has analyzed over 30 different types of human tumors. At the core of its

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Table 1
Summary of the main medical multimodal datasets.

Dataset Name	Modality	Task	Scale	Annotation Type
MIMIC-III	Tabular, text, waveforms	Classification, text generation, clinical prediction	38,597 adult patients, 49,785 hospital admissions, 53,423 ICU stays, 7,870 neonates	Structured clinical records, ICD-9, DRG & CPT codes, free-text clinical notes
MIMIC-IV	Tabular, text	Classification, text generation, clinical prediction	180,733 patients, 431,231 hospital admissions, 73,181 ICU stays	Structured clinical records, ICD-9/10, DRG & HCPCS codes, free-text clinical notes
IU X-Ray	Image, text	Cross-modal retrieval, classification, report generation	3,955 reports, 7,470 images	Manual MeSH and RadLex codes, automated MTI codes, binary labels, free-text radiology reports
MIMIC-CXR	Image, text, tabular	Classification, report generation, cross-modal retrieval, vision-language pre-training (VLP)	65,379 patients, 227,835 imaging studies, 377,110 images	Free-text radiology reports, examination type codes, DICOM metadata
CheXpert	Image, tabular	Classification, pathology detection	65,240 patients, 224,316 chest radiographs	Rule-based diagnostic labels, uncertainty labels, expert consensus annotations
TCGA	Multi-omics, image, tabular	Classification, survival prediction	Thousands of tumor samples, over 30 tumor types	Survival outcomes, diagnostic annotations, tumor subtype labels, molecular profiles
Quilt-1M	Image, text	VLP, cross-modal retrieval	1M image-text pairs	Narrative descriptions, UMLS entities, sub-pathology classes, magnification labels
Harvard-FairVLMed	Image, text, tabular	VLP, classification, fairness evaluation	10,000 image-text pairs	Clinical notes, demographic attributes, disease labels
PMC-OA	Image, text	VLP, cross-modal retrieval	1,646,592 image-text pairs	Figure captions
BIOMEDICA	Image, text	VLP, classification, cross-modal retrieval	24,076,288 image-text pairs, 6,042,494 articles	Captions & figure references, hierarchical concept labels, panel types
MultiCaRe	Image, text, tabular	VLP, classification, image captioning	135,596 images, 96,428 clinical cases	Captions & clinical cases, hierarchical image labels, patient demographics
PrimeKG	Knowledge graph (KG), text	Link prediction, KG representation learning, KGQA	17,080 diseases, 129,375 nodes, 4,050,249 edges	Node labels, edge labels, clinical guidelines

research methodology is multi-omics analysis, which involves the systematic characterization of the same tumor samples at multiple molecular levels. Its data types primarily encompass genomics, epigenomics, transcriptomics, and proteomics data. For a Multimodal RAG, TCGA provides a deep, molecular-level knowledge base that can be retrieved to answer questions about cancer biology, therapeutic responses, and genetic drivers, making it an invaluable knowledge resource in the field of cancer research [150].

Histopathology datasets: Quilt-1M [151] is the largest publicly available histopathology dataset to date, comprising one million image-text pairs. Its data curation process is highly automated, employing a complex pipeline that includes automatic speech recognition (ASR), LLMs, human knowledge databases, and handcrafted algorithms to skillfully integrate data from multiple public sources such as YouTube videos, Twitter, PubMed Open Access articles, and LAION. As a large-scale multimodal resource, Quilt-1M not only covers 18 different histopathology subspecialty classifications in its core dataset, QUILT, but also provides images at various microscope magnifications and rich natural language descriptions that are highly aligned with the image content. It aims to provide a foundational resource for the histopathology domain to advance research in computational

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pathology tasks such as zero-shot classification and cross-modal retrieval.

Ophthalmology datasets: Harvard-FairVLMed [152] is the first medical dataset designed for fairness research in vision-language models. It contains 10,000 patient records for glaucoma diagnosis, with each record including a scanning laser ophthalmoscopy (SLO) fundus image, a clinical note, and detailed demographic attributes (such as age, gender, race, ethnicity, and language). The clinical notes underwent a rigorous de-identification process combining automated tools with manual verification by medical experts. Unlike many datasets that rely on automatically extracted labels, it provides high-quality, gold-standard diagnostic labels, and its clinical notes are rich with non-imaging clinical information beyond image descriptions (such as medication and family history). Harvard-FairVLMed aims to provide a critical foundational resource for the medical AI field to deeply analyze and evaluate the intrinsic biases of vision-language models like CLIP and to promote the development of new algorithms, such as FairCLIP, that balance performance and fairness.

7.1.4. Multi-domain medical datasets

The PMC-OA [153] dataset is a large-scale collection of biomedical image-text pairs derived from the systematic processing of approximately 2.4 million papers from the PubMed Central Open Access subset, ultimately yielding over 1.64 million image-text pairs. The core of its construction process involves filtering for medically relevant images, accurately splitting compound figures from papers into individual subfigures, and then matching them with their corresponding subfigure captions. The medical images in the dataset cover various types, including CT, MRI, X-rays, and microscopy images. The paired text consists of figure captions extracted from the papers, with over 60% being algorithmically aligned subfigure captions that provide fine-grained descriptions closely related to individual subfigures, making it a pre-training resource with high alignment accuracy.

BIOMEDICA [154] is an unprecedentedly large archive of biomedical figures and documents, constructed by processing over 6 million articles from the PubMed Central Open Access subset to create a massive collection of over 24 million figure-text pairs, with each data point equipped with 27 distinct metadata fields. The dataset's construction employed a unique expert-guided annotation process, which first involved unsupervised clustering of the vast number of images, followed by the assignment of fine-grained hierarchical concept labels to the image clusters by a team of multi-domain experts. Its image content is extremely rich and diverse, encompassing not only clinical images from fields like pathology, radiology, and ophthalmology, and microscopy images from cell biology, but also a wide range of non-diagnostic visual materials that are important in scientific literature. The corresponding textual information is also multi-layered, including short core captions, figure mention paragraphs providing more context, and the full text of the papers. BIOMEDICA aims to accelerate the development and evaluation of general-purpose biomedical foundation models by providing a streamable, highly structured, and exceptionally rich resource to serve the broader scientific community.

MultiCaRe [155] is a large-scale, multimodal clinical case dataset sourced from 75,382 open-access case reports published on PubMed Central between 1990 and 2023. This rich dataset comprises 96,428 clinical cases and 135,596 medical images, spanning multiple medical specialties such as oncology, cardiology, surgery, and pathology. During its construction, researchers utilized various tools like Biopython and BeautifulSoup for data collection and combined Spark NLP for Healthcare with manual annotation to extract information from unstructured image captions to generate structured labels. MultiCaRe aims to address the current scarcity of public clinical multimodal datasets, with its data including metadata, clinical cases, images, image labels, and descriptions, providing a valuable resource for the training and fine-tuning of medical vision and language models, as well as for multimodal AI applications.

Precision Medicine Knowledge Graph (PrimeKG) [156] is a large-scale, multimodal knowledge graph built for precision medicine analysis. It integrates 20 high-quality biomedical resources to construct a network containing 129,375 nodes and over 4 million relationships, systematically describing 17,080 diseases. This knowledge graph features a heterogeneous network structure with 10 main types of biological nodes (e.g., diseases, drugs, genes/proteins, anatomy, and phenotypes) and is endowed with multimodal characteristics by augmenting drug and disease nodes with textual descriptions and clinical features from multiple sources (such as DrugBank, Drug Central, and the Mayo Clinic). PrimeKG is designed to provide a comprehensive, computable knowledge base for the field of precision medicine. In a RAG system, it can serve as an external knowledge base to retrieve relevant subgraphs or facts related to a query, which are then fed into the model's context to generate more accurate and evidence-supported responses.

7.2. Evaluation metrics

7.2.1. Metrics for classification and prediction tasks

Standard classification metrics: In predictive analytics, a confusion matrix is a table with two rows and two columns that compares the model's predictions with the actual outcomes, comprising four core elements: (1) True Positive (TP): The number of samples that are actually positive and are correctly predicted as positive by the model; (2) True Negative (TN): The number of samples that are actually negative and are correctly predicted as negative by the model; (3) False Positive (FP): The number of samples that are actually negative but are incorrectly predicted as positive by the model; (4) False Negative (FN): The number of samples that are actually positive but are incorrectly predicted as negative by the model. Accuracy, Precision, Recall, and F1 Score are all standard metrics derived from the confusion matrix [157].

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Accuracy: The proportion of correctly classified samples (including both positive and negative instances) to the total number of samples, calculated as $Accuracy = (TP + TN)/(TP + TN + FP + FN)$. Accuracy measures the overall correctness of the model's predictions. However, this metric has a significant drawback: it can be misleading for imbalanced datasets (e.g., in the case of rare diseases).

Precision: The proportion of true positive samples among all samples predicted as positive by the model, calculated as $Precision = TP/(TP + FP)$. Precision measures how reliable the model's positive predictions are. High precision implies that the vast majority of samples predicted as positive are indeed positive, thus avoiding an excessive number of false alarms.

Recall: The proportion of correctly identified positive instances among all actual positive instances, also known as Sensitivity, calculated as $Recall = TP/(TP + FN)$. Recall measures the model's ability to find all positive instances. This metric is crucial in medical research where the goal is to minimize missed detections. However, striving for an excessively high recall may lead to an increase in FP, thereby affecting precision.

F1 Score: The harmonic mean of Precision and Recall, calculated as $F1 = 2 \cdot Precision \times Recall/(Precision + Recall)$. The F1 Score seeks a balance between precision and recall, and it penalizes models that exhibit extreme differences between the two metrics.

Threshold-independent metrics: Receiver Operating Characteristics (ROC) Plot illustrates the trade-off between sensitivity (True Positive Rate) and specificity (1 - False Positive Rate) by calculating these metrics across all possible classification thresholds. Its x-axis represents the False Positive Rate (FPR), and the y-axis represents the True Positive Rate (TPR). In an ROC plot, a diagonal line from (0,0) to (1,1) serves as the performance baseline for a random classifier. The precision-recall curve (PRC) plot shows the relationship between precision and recall. Its x-axis represents recall, and its y-axis represents precision. Unlike the ROC plot, the PRC plot's baseline is not fixed; it moves depending on the ratio of positives (P) to negatives (N) in the dataset, with the baseline being at $y = P/(P + N)$.

AUROC: A single numerical value derived by calculating the total area under the ROC curve, which is convenient for comparing the overall performance of multiple classifiers. This value typically ranges from 0.5 to 1.0, where 0.5 represents the performance of a random classifier and 1.0 represents a perfect classifier. However, the ROC curve, and by extension its AUROC score, is insensitive to class imbalance in the dataset. This can mask significant differences in a classifier's real-world performance when class distribution is skewed, potentially leading to deceptive conclusions.

AUPRC: The value calculated from the total area under the PRC plot, which is also effective for comparing the performance of multiple classifiers. Unlike AUROC, the AUPRC for a random classifier is not a fixed 0.5; it is equal to the PRC baseline, which corresponds to the proportion of positives in the dataset, $P/(P + N)$. The PRC plot and its AUPRC score are highly sensitive to the dataset's class distribution. As a dataset shifts from balanced to imbalanced, the AUPRC changes accordingly. Because it directly evaluates the fraction of true positives among positive predictions (i.e., precision), it provides a more accurate and intuitive interpretation of a classifier's practical performance [158].

7.2.2. Metrics for generation tasks

Common text metrics: BLEU- This metric was initially proposed for machine translation, evaluating text quality by computing n-gram precision between candidate texts and reference texts, supplemented by a brevity penalty [159]. The metric is simple to compute, language-independent, and therefore widely used in natural language generation tasks. However, BLEU shows poor correlation with human judgments at the sentence level and, due to its reliance on literal word matching, struggles to effectively capture semantic similarity.

ROUGE- A set of metrics for automatically evaluating the quality of a summary by comparing it with ideal summaries created by humans. Unlike the precision-based BLEU, the core of ROUGE is recall. Among its variants, ROUGE-N measures the degree of overlap by calculating the recall of n-grams between the candidate and reference summaries, while ROUGE-L measures the longest common subsequence to better capture sentence-level structure by rewarding in-sequence word matches [160]. However, this metric is still limited by its reliance on surface-level lexical overlap.

METEOR- This metric is based on unigram alignment between the candidate and reference texts. It matches not only exact words but also word variants and synonyms through stemming and external knowledge bases like WordNet. METEOR combines unigram precision and recall and introduces a fragmentation penalty, which is calculated based on the number of matched chunks to reward more fluently ordered sentences. METEOR generally shows a higher correlation with human judgments than BLEU [161].

BERTScore- This metric addresses the limitations of n-gram-based metrics by computing similarity using contextual embeddings from BERT. It calculates precision, recall, and F1 scores based on the cosine similarity between token embeddings in the candidate and reference texts. By evaluating similarity at a semantic level rather than at the lexical surface form, BERTScore can effectively handle paraphrases and capture long-range dependencies, thus aligning more closely with human judgments across various tasks than traditional metrics [162].

Clinical factuality metrics: F1-CheXbert- This metric utilizes the CheXbert labeler as a proxy for radiologists to evaluate the factual accuracy of generated reports. During evaluation, both the generated and reference reports are fed into CheXbert to extract 14 clinical observations. For each observation, CheXbert outputs its state (e.g., positive, negative, uncertain, or blank). An F1 score is then calculated based on the consistency of these labels between the generated and reference reports [163]. The key advantage of this

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method is its focus on the consistency of high-level clinical information, rather than on traditional lexical similarity.

F1-RadGraph- This metric provides a fine-grained evaluation of factual correctness by leveraging the RadGraph framework. RadGraph comprises a knowledge graph-based information extraction schema that parses radiology reports into a knowledge graph of clinical entities and their relations. The F1-RadGraph score is calculated based on the F1 overlap of entity-relation triplets between the generated and reference reports [164]. It is considered a robust metric for factual correctness as it assesses the identification of key clinical entities, their structural relationships, and their semantics.

LLM-as-a-Judge- Recently proposed as an emerging evaluation paradigm, it offers a feasible and scalable method that employs LLMs as evaluation tools to assess model-generated outputs based on a series of predefined evaluation criteria, instructions, and context [165]. This paradigm can be extended to medical Multimodal RAG systems, offering unique value in assessing the comprehensive capabilities of the entire system. For instance, LLM-as-a-Judge can be used to evaluate whether the retriever has found highly relevant multimodal medical knowledge for a given query, whether the generator faithfully utilizes the retrieved context to construct the final answer, and the quality of the final generated response.

8. Performance of representative frameworks

While the representative studies mentioned previously all leverage Multimodal RAG techniques to enhance overall system performance, they exhibit significant diversity in their targeted clinical tasks, adopted datasets, and evaluation methodologies. This section aims to provide a comprehensive performance evaluation and comparison of representative frameworks in the field of medical Multimodal RAG.

8.1. Performance in Med-VQA and report generation tasks

FactMM-RAG focuses on the automated generation of accurate radiology reports. The framework was evaluated on two chest radiology benchmark datasets: MIMIC-CXR and CheXpert. Its evaluation system not only incorporates traditional natural language generation metrics such as ROUGE-L and BERTScore but also introduces metrics specifically designed for clinical accuracy: F1CheXbert and F1RadGraph. Experimental results demonstrate that FactMM-RAG outperforms SOTA retrievers, with its F1CheXbert score increasing by up to 6.5% and the F1RadGraph score by 2% on the MIMIC-CXR dataset. In zero-shot evaluations on the CheXpert dataset, these two metrics also showed improvements of 2% and 1.2%, respectively, compared to the SOTA retriever.

RULE and MMed-RAG both demonstrate significant improvements in factual accuracy in both medical VQA and report generation tasks. RULE was evaluated on these two tasks across three datasets covering radiology and ophthalmology (MIMIC-CXR, IU-Xray, and Harvard-FairVLMed). For the medical VQA task, RULE showed the best overall performance compared to baseline methods in terms of Accuracy, Precision, Recall, and F1 Score. For the report generation task, RULE outperformed baseline methods on all metrics, including BLEU Score, ROUGE-L, and METEOR. Across all datasets and both tasks, RULE improved the average accuracy by 47.4%. MMed-RAG's evaluation of the medical VQA task further included pathology datasets (PMC-OA and Quilt-1M) and used Accuracy, F1 Score, and AUROC as evaluation metrics; for the report generation task, it used the same evaluation metrics as RULE. Overall, MMed-RAG outperformed all baselines (including RULE and FactMM-RAG) on nearly all metrics and datasets.

V-RAG is dedicated to reducing the hallucination phenomenon in Med-MLLMs through entity probing and subsequent report correction. This framework was validated on the MIMIC-CXR and MultiCaRe datasets. Its performance evaluation primarily relied on Precision, Recall, and F1 Score for the entity probing task and the RadGraph-F1 score for the report generation task. In the entity probing task, V-RAG (fine-tuned) achieved F1 scores of 0.751 and 0.940 on MIMIC-CXR and MultiCaRe, respectively, significantly outperforming the text-only RAG baseline model. More importantly, correcting the initially generated reports using the entity probing results from V-RAG led to a relative improvement of 19% in the simple and partial RadGraph-F1 scores.

The multimodal multi-agent framework proposed by Yi et al. [142] focuses on the radiology report generation task, enhancing report quality through a sophisticated architectural design. The framework was evaluated on the MIMIC-CXR and IU X-ray datasets. In addition to using standard metrics like BLEU, ROUGE, METEOR, and BERTScore, it also innovatively introduced the "LLM-as-a-Judge" paradigm, utilizing the Claude 3 Opus model to score the generated reports on a scale of 1-10 across five dimensions, including diagnostic accuracy, conciseness, and consistency with original reports. Compared to the single-agent baseline LLaVA-Med, this framework achieved significant improvements across all metrics except for consistency. For instance, the BERTScore increased from 0.8617 to 0.8819, and the diagnostic accuracy score rose from 7.78 to 8.26.

8.2. Performance in clinical classification and diagnosis tasks

The EMERGE framework aims to leverage multimodal EHR data for clinical prediction. It underwent extensive experimentation on two real-world EHR datasets, MIMIC-III and MIMIC-IV, for two binary classification tasks: in-hospital mortality prediction and 30-day readmission prediction. Performance evaluation adopted widely used clinical metrics, including AUROC and AUPRC, as well as the min(+P, Se) metric, which was included to ensure high precision and high sensitivity. The experimental results show that EMERGE consistently outperformed baseline models across all tasks and datasets. For instance, its AUROC reached 86.25 in the mortality prediction task on MIMIC-III

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and 80.61 in the readmission prediction task on MIMIC-IV, demonstrating its practicality in real-world applications.

The XLR-KGDD framework utilizes multimodal clinical information for explainable disease diagnosis. It was evaluated on the MIMIC-Eye dataset for the diagnosis of Congestive Heart Failure (CHF) and Urinary Tract Infection (UTI). Performance metrics included accuracy, precision, recall, F1-score, AUC of ROC, and AUC of PR. When using the fine-tuned LLaMA-2 model, the XLR-KGDD framework outperformed other LLM-based disease diagnosis studies, including Health-LLM, Gemini Model, and GPT-3.5, in terms of accuracy, recall, and F1-score. With the fine-tuned LLaMA-3 model, the AUC of ROC for Congestive Heart Failure diagnosis was 0.88 and the AUC of PR was 0.91; for Urinary Tract Infection diagnosis, the AUC of ROC was 0.90 and the AUC of PR was 0.83.

The RAICL framework is designed to enhance the performance of MLLMs on disease classification tasks through retrieval-augmented in-context learning, particularly in few-shot scenarios. It was evaluated on two distinct datasets: the histopathology image dataset TCGA and the chest X-ray dataset IU-CXR. Evaluation metrics included overall accuracy, as well as micro-averaged and macro-averaged precision, recall, and F1 score. The most notable achievement of RAICL is its ability to deliver significant performance improvements without requiring additional model fine-tuning. For example, on the TCGA dataset, it increased the accuracy of the Qwen-2.5-VL-7B model from 0.7854 to 0.8368 and, more strikingly, dramatically boosted the macro-averaged F1 score from a very low 0.1191 to 0.8454.

9. Future directions

Despite remarkable progress, medical Multimodal RAG remains in rapid evolution. This section outlines critical directions for future research, including enhancing cross-modal reasoning and exploring few-shot and zero-shot learning for data scarcity. We also emphasize the urgency of establishing standardized evaluation frameworks and discuss ethical considerations for clinical deployment, alongside the potential of integrating RAG with other emerging AI technologies.

9.1. Enhanced cross-modal understanding and reasoning

A core technical obstacle for multimodal artificial intelligence in the medical domain is the semantic gap and the challenge of alignment between modalities. The prevailing strategy to address this issue is learning a shared latent embedding space. The central idea is to project concepts from different modalities into a unified, high-dimensional vector space, wherein semantically similar concepts are brought closer together. Researchers are also exploring more novel architectures to solve the fundamental problem of cross-modal alignment. For example, M^3Bind, proposed by Liu et al. [166], is an innovative pre-training framework capable of aligning multiple medical imaging modalities (such as X-rays and ECGs) without direct paired data by using text as a universal bridge. The CAT framework, proposed by Tang et al. [167], focuses on automated medical report generation. Prior to the main generation task, its Keyword-Region Pre-alignment module leverages an attention mechanism to align key medical terms from the report with relevant regions in the medical image. Subsequently, a retriever is used to find similar and information-rich reports from a knowledge base, thereby enhancing the encoder's contextual information.

Even when the alignment problem is addressed, the reasoning process of many current models remains an opaque black box, significantly hindering their clinical trustworthiness. Consequently, a key future direction is to render the reasoning process explicit and procedural. MCoT offers a powerful theoretical framework to achieve this. MCoT extends the mature CoT paradigm from language models to multimodal scenarios, enabling the model to generate a series of intermediate reasoning steps that combine visual evidence with textual knowledge, ultimately leading to a conclusion [168]. This technology allows clinicians to review, validate, and even correct each step of the model's reasoning, thereby improving the trustworthiness of the system.

To handle more complex clinical scenarios, future RAG systems will adopt more advanced architectural designs that simulate the collaborative and reflective processes of human experts. An emerging method is to build structured, hierarchical reasoning. This approach first utilizes multiple models, fine-tuned in various ways, to generate preliminary diagnostic opinions. Subsequently, a top-level reasoning layer synthesizes, adjudicates, and corrects these diverse outputs to arrive at a final conclusion that is more robust and reliable than that of any single model. This method holds significant clinical promise as it mimics the workflow of human physicians who discuss difficult cases in consultation to reduce misdiagnosis. Another forward-looking paradigm is Agentic RAG. In this system, autonomous AI agents are integrated into the RAG process, capable of performing multi-step reasoning, reflecting on acquired information, and actively deciding whether and how to query external knowledge bases based on task complexity and uncertainty [169]. Future developments in medical Agentic RAG can draw methodological inspiration from other strategy-intensive disciplines. For example, the TANS-Agent framework demonstrates how grounding multi-agent systems in explicitly structured semantics can overcome the limitations of unstructured data, providing a robust reference for building more precise and explainable reasoning pipelines in complex clinical environments [170].

9.2. Multimodal RAG under few-shot and zero-shot learning

In the field of medical AI, a significant long-standing obstacle is the scarcity of high-quality annotated data, particularly for rare diseases, specific subspecialties, or emerging imaging technologies. Against this backdrop, the RAG paradigm is emerging as a key enabling technology for achieving efficient learning in data-scarce environments. Developing systems that can learn effectively under limited

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data conditions—that is, possessing few-shot and zero-shot learning capabilities—has become a core direction for the future development of medical Multimodal RAG.

Few-shot segmentation is a typical application scenario of data-efficient learning, which aims to enable a model to segment new, unseen classes using only a very small number of annotated samples. This is highly valuable in medical image analysis, for instance, enabling the rapid deployment of a segmentation model for a rare type of tumor. Zhao et al. [171] proposed an innovative method that combines the RAG mechanism with foundation models. When segmenting a new target image, the system first utilizes the powerful visual feature extraction capabilities of DINOv2 to retrieve the most similar samples from a small, labeled support set database. These retrieved samples, including both the images and their corresponding segmentation masks, are then fed as contextual information into the memory attention mechanism of SAM 2. SAM 2 leverages these memories as dynamic prompts to guide the segmentation of the new image. This entire process requires no retraining or fine-tuning of the foundation models. Instead, the RAG mechanism dynamically injects task-relevant prior knowledge into the general-purpose foundation model, thereby achieving strong generalization capabilities across different medical modalities.

Zero-shot learning aims to maximize data efficiency, with its classic definition being the ability of a model to recognize classes unseen during the training phase. In the context of RAG systems, this typically refers to the model's ability to generate responses by relying on its powerful pre-trained generalization and reasoning capabilities, combined with context from a retrieval module, all without any task-specific fine-tuning [172]. For example, FilterRAG is a framework designed for VQA that incorporates zero-shot learning as a core principle. The method integrates BLIP-VQA with RAG to ground answers in external knowledge sources like Wikipedia. This aims to enhance model robustness and ensure generalization to unseen tasks and Out-of-Distribution (OOD) scenarios [173]. Empowering medical Multimodal RAG systems with zero-shot capabilities, thereby enabling good generalization on unseen domains or tasks, represents a cutting-edge and challenging research direction for the future.

9.3. Establishing a standardized evaluation framework

Although medical Multimodal RAG demonstrates immense application potential, its development faces a critical bottleneck: the lack of a unified, standardized evaluation framework that can reflect real-world clinical value. Existing medical Multimodal RAG frameworks are often designed for specific research purposes, using varied datasets and metrics. This makes it difficult to directly compare results across different studies, thereby exacerbating the fragmentation of the evaluation landscape.

A unified evaluation standard requires the support of general-purpose, representative, and high-quality datasets. Constructing large-scale, diverse, and expertly annotated medical datasets capable of supporting the training and evaluation of complex multimodal models is extremely challenging, especially in the field of rare diseases where high-quality data is exceptionally scarce [174]. Many current medical benchmarks are derived from academic literature or standardized exams. These are far removed from the real-world clinical scenarios of processing unstructured, noisy, and contextually complex information from patient records and thus fail to truly reflect the complexities of clinical practice [175]. Furthermore, many datasets and applications focus on a single medical domain or a limited number of task types, which greatly restricts the assessment of a system's cross-domain generalization and comprehensive reasoning capabilities. Future work needs to focus on constructing common, high-quality datasets that cover a wide range of data modalities and span various clinical tasks. This will enable a more comprehensive and realistic assessment of framework capabilities.

In terms of evaluation metrics, traditional metrics from the general natural language processing domain, such as BLEU and ROUGE, are fundamentally based on surface-level text matching via n-gram overlap. These metrics fail to comprehend the deep semantics of medical terminology and cannot adequately judge the factual accuracy, logical coherence, and clinical relevance of a generated diagnosis [176]. Even more clinically relevant metrics, such as F1-CheXpert and F1-RadGraph, are often limited by their lack of interpretability [177]. Innovative evaluation methods and metrics with greater clinical utility should continue to be developed in the future. For example, in report evaluation, RaTEScore proposed by Zhao et al. [178] focuses on the medical entity level rather than the word level, enabling it to align more closely with human clinical judgment than traditional metrics. Similarly, GEMA-Score, proposed by Zhang et al. [179], utilizes a multi-agent framework to assess multiple aspects of a report, thereby yielding more fine-grained and explainable evaluation results.

9.4. Ethical and deployment considerations for clinical applications

Although Multimodal RAG systems show broad potential in the medical field, their path from the laboratory to real-world clinical practice is fraught with complex societal challenges involving ethics, regulations, and deployment barriers.

The primary ethical concern is preventing patient harm from the AI's erroneous or misleading information. Clear guidelines must be established to prevent over-reliance on AI and to emphasize its role as a decision-support tool rather than a decision-maker. Any integration of AI systems into clinical workflows must be supervised by qualified healthcare professionals [180]. Determining how to assign liability when an AI system errs and causes adverse outcomes is a pressing legal and ethical dilemma.

The development and application of any AI system must strictly adhere to data protection regulations. Patient data

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constitutes extremely sensitive personal information, and the retrieval datasets that RAG systems rely on may inadvertently contain such sensitive patient information. Therefore, the use of patient information must be based on the explicit informed consent of patients, and effective data anonymization techniques and secure data management protocols must be implemented to ensure privacy and data security [181, 182].

If an AI model is trained on biased data, it risks entrenching or even amplifying existing real-world healthcare inequalities. For example, if training data is primarily sourced from a specific population, the model's performance on other populations may decline, leading to biased diagnostic and treatment recommendations [183]. Furthermore, most existing datasets and models exhibit a significant linguistic bias (predominantly English-centric), limiting their equitable application on a global scale. Developing techniques to mitigate or eliminate model bias to enhance system fairness is a crucial direction for future efforts.

Transforming a prototype system into a clinically viable product also requires overcoming numerous practical deployment hurdles. The EHR systems, imaging equipment, and data formats can vary widely across different hospitals, leading to high data heterogeneity. This necessitates cross-departmental collaboration to handle highly heterogeneous and incomplete real-world medical datasets, ensuring that the AI system can seamlessly integrate with existing hospital information systems [184]. Moreover, real-world deployment demands rigorous protocols for the continuous maintenance and updating of external knowledge bases. Medical knowledge and clinical guidelines evolve rapidly; thus, relying on a static or outdated retrieval corpus poses a significant clinical risk, necessitating automated pipelines for real-time knowledge ingestion. Finally, the industry currently lacks unified, authoritative validation frameworks and evaluation standards. Before models can be deployed on a large scale, their safety, reliability, and clinical efficacy must be comprehensively evaluated through standardized processes.

9.5. Integration with other AI technologies

Traditional RAG systems often employ static retrieval mechanisms, such as those based on vector similarity search. For complex, multi-hop clinical questions that require an iterative information-gathering process, this single-pass retrieval strategy may not be optimal. RL can be used to train a policy for the RAG retriever. An RL agent learns to dynamically adjust its retrieval behavior through interaction with its environment. Based on the current query and the information already retrieved, it selects the next retrieval action to maximize a predefined reward signal. This transforms the retrieval process from a single-step search into a multi-step, adaptive reasoning process [185]. By dynamically optimizing its information-gathering path, the model can ultimately ensure retrieval relevance while significantly enhancing the accuracy of task completion on complex problems.

Due to its high sensitivity, medical data is often isolated within individual healthcare institutions, creating data silos. Federated learning (FL) offers an innovative solution to this dilemma. It is a distributed, privacy-preserving machine learning framework. In a typical FL process, a global model is distributed to participating institutions (e.g., hospitals), each of which trains the model on its local, private data. Afterward, updated model parameters (such as gradients) are sent back anonymously to a central server for aggregation, which in turn updates the global model [186]. A promising future direction involves the organic integration of Multimodal RAG systems with FL. Within a feasible technical architecture, such a system could employ a Federated Retrieval mechanism to securely query private knowledge bases distributed across a consortium of hospitals, leveraging technologies such as homomorphic encryption [187]. Furthermore, the generator can be fine-tuned through personalized FL, enabling each institution to train a personalized model specifically adapted to its local data distribution and available modalities, while simultaneously benefiting from the shared knowledge derived from collaborative training [188, 189].

10. Conclusion

By integrating diverse medical data sources, medical Multimodal RAG has demonstrated significant potential in tasks such as clinical diagnostic assistance, medical question answering, and automated report generation, significantly enhancing the accuracy and factual consistency of models. By incorporating verifiable external knowledge, this paradigm effectively mitigates the issues of outdated knowledge and hallucination in large models and serves as a key pillar for building the next generation of highly reliable, general-purpose intelligent healthcare systems. Future research will focus on integrating Multimodal RAG with other AI technologies to achieve more refined cross-modal understanding and complex reasoning, explore data-efficient learning methods, and address a range of technical and societal challenges during deployment.

CRediT authorship contribution statement

Ruipeng Wang: Writing – original draft, Writing – review and editing, Visualization, Investigation, Data curation, Formal analysis, Conceptualization. **Jian Wang:** Writing – review and editing, Investigation. **Yan Huang:** Writing – review and editing. **Haiyang Guo:** Writing – review and editing, Investigation. **Junjie Pang:** Writing – review and editing, Supervision, Visualization, Conceptualization.

Acknowledgements

This work was supported by the Scientific Research Project of the General Administration of Customs of the People's Republic of China (Grant No. 2024HK208).

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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